

Quantum minimum description of density matrices

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We study a variant of Schumacher compression. The task asks for the minimal memory cost for compressing many copies of a density matrix with known spectrum and unknown eigenbasis, without preserving its purification. For fixed dimension and distinct positive eigenvalues, we obtain the cost through its additive constant. Achievability uses a generalization of Werner’s cloning map to $GL(d, \mathbb{C})$ irreducible representations, with finite trace-distance bounds controlled by highest-weight differences and row gaps. The matching converse follows from a quantitative form of Koashi–Imoto incompressibility for irreducible group orbits under a spectral-gap assumption. We also identify the relation between this quantum minimum description length and universal lossless coding overhead, and its connection to free entropy is explained in a companion letter [1]. We provide a [Lean certificate](#) for our proofs.

1 Introduction

Schumacher compression characterizes the memory needed to preserve quantum information together with its correlations with a reference [2]. A different problem arises when one only needs to reconstruct many identical copies of a state whose eigenbasis is unknown. The memory must retain this unknown basis information, but need not preserve a purification. We determine the optimal memory cost through its additive constant and construct a protocol attaining it.

The task is to compress identical and independent copies of a density matrix whose spectrum $x = (x_1, \dots, x_d)$ is known but whose eigenbasis is not. Fix a reference state ρ with this spectrum and write $\rho_g := U(g)\rho U(g)^\dagger$ for $g \in U(d)$. Let M_n be the sole retained memory, including any classical register, and write $|M_n| := \log \dim M_n$; all logarithms are base two. The encoder $\mathcal{E}_n : \mathcal{B}((\mathbb{C}^d)^{\otimes n}) \rightarrow \mathcal{B}(M_n)$ and decoder $\mathcal{D}_n : \mathcal{B}(M_n) \rightarrow \mathcal{B}((\mathbb{C}^d)^{\otimes n})$ are completely positive trace-preserving maps, which may depend on n and x but not on g . They form a $(|M_n|, \delta)$ -code if

$$\sup_{g \in U(d)} \frac{1}{2} \|\rho_g^{\otimes n} - \mathcal{D}_n \circ \mathcal{E}_n(\rho_g^{\otimes n})\|_1 \leq \delta. \quad (1)$$

We seek the smallest asymptotic memory size among sequences with $\delta_n \rightarrow 0$. We call this cost the *quantum minimum description length* (QMDL) of the known-spectrum orbit family. A completely known state can be regenerated without storing its input; the

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unknown eigenbasis makes the present task nontrivial. For a nontrivial orbit, a purely classical strategy such as tomography followed by state preparation cannot achieve vanishing global error [3]. The cost $|M_n|$ is the log dimension, rather than the rounded number $\lceil \log \dim M_n \rceil$ of physical qubits; this distinction matters for the additive constant.

Theorem 1 (Optimal memory cost). *Fix d and a rank- r density matrix with distinct positive eigenvalues $x_1 > \dots > x_r > 0$. Define*

$$L_{d,r}(n, x) := \frac{r(2d - r - 1)}{2} \log n + \sum_{1 \leq i < j \leq r} \log(x_i - x_j) + (d - r) \sum_{i=1}^r \log x_i - \sum_{k=d-r}^{d-1} \log k!. \quad (2)$$

There exists a sequence of codes with $|M_n| = L_{d,r}(n, x) + o(1)$ and $\delta_n = O(\log n / \sqrt{n})$. Every sequence of codes with $\delta_n \rightarrow 0$ satisfies

$$\liminf_{n \rightarrow \infty} [|M_n| - L_{d,r}(n, x)] \geq 0.$$

The lower bound also holds when only the Haar-average reconstruction error tends to zero.

Theorem 1 combines the achievability and converse results proved in Sections 4 and 5. The dimension and spectrum are fixed throughout these limits; no uniform claim is made as positive eigenvalues collide or approach zero. The theorem asserts a sequence of codes that achieves this expansion with vanishing error, rather than the same expansion for every prescribed error schedule.

Earlier work established the leading memory cost [4, 3]. For qubits, the protocol of Yang, Chiribella, and Hayashi (YCH) already has an achievable spectrum-dependent constant. We prove the matching converse through that constant and extend the construction to qudits. The coefficient of $\log n$ counts half the real dimension of the orbit, while the additive term resolves the eigenvalue dependence.

The construction uses covariant channels between $\mathrm{GL}(d, \mathbb{C})$ irreducible representation spaces. Their description by normalized Choi projectors is established representation-theoretic machinery [5, 6, 7]. We select the Parthasarathy–Ranga Rao–Varadarajan (PRV) component, obtaining channels that extend Werner’s symmetric-power cloner [8]. For $d \geq 2$ and partitions μ, ν with at most r rows and dominant difference $\omega = \nu - \mu$, Theorem 2 gives the finite trace-distance bound $C_{d,x} D / (b_\mu + 1)$ in both directions, where $D = \|\omega\|_1$ and b_μ is the minimum of the first $\min(r, d - 1)$ adjacent-row gaps of μ . A padded target representation stores all typical Schur blocks in the same memory space. For each such block and this target, $b_\mu = \Omega(n)$ and $D = O(\sqrt{n} \log n)$, giving error $O(\log n / \sqrt{n})$.

The converse uses a quantitative incompressibility bound for irreducible compact-group orbits (Proposition 4). If the orbit state on $\mathcal{H}$ has a simple largest eigenvalue separated from the remaining spectrum by γ , then any code with Haar-average error $\bar{\delta}$ requires $\dim M \geq \dim \mathcal{H}(1 - \bar{\delta}/\gamma)$. This bound applies to arbitrary encoding and decoding channels. A uniform lower bound on γ for the representation states lets us retain the additive memory constant when transferring the bound to the original n -copy family.

Universal lossless quantum data compression uses one code for every state in the source family. Even with a known spectrum, the unknown eigenbasis increases the required average code length. We show that the smallest worst-case excess in ideal average length above $nS(\rho)$ is $L_{d,r}(n, x) - s_{\mathrm{blk}}(x) + o(1)$. Thus QMDL also quantifies this coding overhead, up to an explicit entropy correction: $s_{\mathrm{blk}}(x)$ is the limiting entropy of the irreducible representation state in a Schur block. Section 6 identifies the optimal code state; Appendix B evaluates the additional classical penalty when the spectrum is unknown.

The spectral terms in the memory cost also have a geometric interpretation. For the rank- r states in Theorem 1, the unitary orbit $\Omega = \{U\rho U^\dagger\}$ has real dimension $r(2d-r-1)$. Its volume in the induced Hilbert–Schmidt metric is

$$\text{Vol}_{\text{HS}}(\Omega) = c_{d,r} \prod_{1 \leq i < j \leq r} (x_i - x_j)^2 \prod_{i=1}^r x_i^{2(d-r)},$$

where $c_{d,r}$ depends only on d, r . One half of its logarithm therefore reproduces the spectral terms in $L_{d,r}(n, x)$. Our companion letter [1] makes the comparison precise using a Hilbert–Schmidt tube-volume definition of physical free entropy and relates it to Voiculescu’s microstate free entropy [9, 10, 11]. Asymptotically, the optimal memory cost equals one half of that finite-dimensional quantity at resolution n^{-1} , up to an explicit offset depending only on d, r . This is a comparison between geometric volume and representation dimension; the resolution belongs to the geometric counting definition.

Section 2.1 reviews prior compression results, and Section 2.2 recalls the qubit protocol. Section 3 defines the PRV channels and their structural properties. Sections 4 and 5 prove the optimal memory theorem. Section 6 relates QMDL to the extra code length required by universal lossless compression. The technical trace-distance estimate is proved in Section 7.

The proofs of our main theorems are formalized in Lean and Comparator checked. See the [Lean Certificate](#) and an accompanying [Proof Wiki guide](#).

2 Background

2.1 Related work

Quantum minimum description length connects the classical idea of description length [12] with compression of repeated quantum states. Plesch and Bužek [13] gave an $O(\log n)$ -qubit compression scheme for n identical pure qubits. The reduction uses their support on the symmetric subspace, whose dimension grows polynomially with n . Rozema et al. implemented the compression of three pure qubits into two [14]. For mixed states, the algebraic basis of exact compression is given by Koashi and Imoto [15, 16].

The YCH scheme [4] established the leading memory cost for mixed qubits and an achievable spectrum-dependent constant. It also underlies the “quantum stopwatch” protocols of [17, 18]. Hayashi and Tan [19] determined the optimal compression of diagonal states. For qudits, the $O(\log n)$ scheme of [20] had a suboptimal prefactor; Yang et al. [3] subsequently obtained the optimal leading cost, with roughly $\frac{1}{2} \log n$ per unknown parameter. Their memory combines quantum and classical registers; a purely classical memory does not suffice for unknown-basis families.

Further work treats states generated by shallow circuits with both n and d growing [21], and experimental implementations appear in [22, 23]. Here we determine the optimal additive constant for fixed spectra with distinct positive eigenvalues and extend the YCH construction to qudits.

2.2 The YCH qubit protocol

The YCH protocol [4] compresses n copies of $\rho = U(g)\rho_0 U(g)^\dagger$, where $g \in \text{U}(2)$ is unknown and $\rho_0 = \text{diag}(p, 1-p)$ with $1/2 < p \leq 1$. At $p = 1/2$ the orbit is a single known state and needs no memory.

The Schur transform resolves the permutation symmetry:

$$V_{\text{Schur}} : (\mathbb{C}^2)^{\otimes n} \rightarrow \bigoplus_{J=j_{\min}}^{n/2} \mathcal{H}_J \otimes \mathcal{M}_J \quad (3)$$

Here $\mathcal{H}_J$ is the $\text{GL}(2, \mathbb{C})$ irrep of dimension $2J+1$ labeled by $(n/2+J, n/2-J)$, and $\mathcal{M}_J$ is the corresponding symmetric-group irrep. The allowed spins start at $j_{\min} = 0$ for even n and $j_{\min} = 1/2$ for odd n .

Applying the Schur transform to the input state block-diagonalizes it:

$$V_{\text{Schur}} \rho^{\otimes n} V_{\text{Schur}}^\dagger = \bigoplus_{J=j_{\min}}^{n/2} q_J \rho_{g,J} \otimes \tau_J \quad (4)$$

where $\rho_{g,J} := U_J(g) \rho_J U_J(g)^\dagger$ and τ_J is maximally mixed on $\mathcal{M}_J$. Writing $q := 1 - p$, the probability of the J sector is

$$q_J = \dim \mathcal{M}_J \cdot s_{(\frac{n}{2}+J, \frac{n}{2}-J)}(p, q) = \frac{2J+1}{\frac{n}{2}+J+1} \binom{n}{\frac{n}{2}-J} \sum_{j=\frac{n}{2}-J}^{\frac{n}{2}+J} p^j q^{n-j}, \quad (5)$$

where s_λ is the Schur polynomial and $\dim \mathcal{M}_J = \frac{2J+1}{n/2+J+1} \binom{n}{n/2-J}$.

The protocol stores every measured sector in one target irrep:

Qubit Compression Scheme (YCH [4])

Encoding:

1. **Irrep Measurement:** Perform a non-demolition measurement of the quantum number J , preserving the quantum information within the associated subspace $\mathcal{H}_J \otimes \mathcal{M}_J$.
2. **Cloning:** Choose J^* to be the allowed spin in $\{j_{\min}, j_{\min} + 1, \dots, n/2\}$ nearest the concentration center $(p - 1/2)(n + 1)$. This need not be the exact mode of q_J ; at $p = 1$ it gives $J^* = n/2$. Discard the multiplicity register $\mathcal{M}_J$ and apply the cloning channel $\mathcal{C}_{J \rightarrow J^*}$ to map the remaining state $\rho_{g,J}$ to a fixed irrep J^* . Store this output state into a quantum memory of dimension $2J^* + 1$.

Decoding:

1. **Irrep Sampling:** Pick a target value K at random from the distribution q_K .
2. **Cloning:** Apply the optimal cloning channel $\mathcal{C}_{J^* \rightarrow K}$ to the state in the quantum memory.
3. **State Reconstruction:** Append a new multiplicity register initialized in the maximally mixed state τ_K , and perform the inverse Schur transform.

Only the irrep state carries the unknown eigenbasis. The multiplicity register can be discarded and regenerated, while the known distribution q_J can be resampled. Cloning into a single target irrep removes the need to retain J ; concentration near J^* controls the error.

The channel uses Werner's optimal approximate cloner from n to $m \geq n$ identical pure states [8]:

$$\mathcal{C}_{n \rightarrow m} : \text{Sym}^n(\mathbb{C}^d) \rightarrow \text{Sym}^m(\mathbb{C}^d), \quad \rho \mapsto \frac{d_{\text{Sym}^n(\mathbb{C}^d)}}{d_{\text{Sym}^m(\mathbb{C}^d)}} V^\dagger(\rho \otimes \mathbb{I}_{m-n}) V \quad (6)$$

where $\mathbb{I}_{m-n}$ is the identity on $\text{Sym}^{m-n}(\mathbb{C}^d)$, and V embeds $\text{Sym}^m(\mathbb{C}^d)$ isometrically into $\text{Sym}^n(\mathbb{C}^d) \otimes \text{Sym}^{m-n}(\mathbb{C}^d)$.¹ It is fixed up to a phase by

$$VU(g)^{\otimes m} = (U(g)^{\otimes n} \otimes U(g)^{\otimes(m-n)})V, \quad g \in U(d). \quad (7)$$

As an $SU(2)$ representation, $\mathcal{H}_J \simeq \text{Sym}^{2J}(\mathbb{C}^2)$; determinant factors cancel under conjugation. Werner's cloner therefore maps spin J to spin $K \geq J$, while partial trace handles $J > K$:

$$\mathcal{C}_{J \rightarrow K}(\rho_J) := \begin{cases} \frac{2J+1}{2K+1} V^\dagger(\rho_J \otimes \mathbb{I}_{2K-2J}) V, & J \leq K \\ \text{Tr}_{2J-2K}(\rho_J), & J > K \end{cases}. \quad (8)$$

The intertwining property of V and unitary invariance of the auxiliary identity make this channel covariant:

$$\mathcal{C}_{J \rightarrow K}(U_J(g)\rho_J U_J^\dagger(g)) = U_K(g)\mathcal{C}_{J \rightarrow K}(\rho_J)U_K^\dagger(g). \quad (9)$$

YCH demonstrated that this compression scheme achieves asymptotically vanishing error. Since $J^* = (p - 1/2)(n + 1) + O(1)$, its achievable memory cost is

$$|M_n| = \log(2J^* + 1) = \log n + \log(2p - 1) + o(1). \quad (10)$$

3 Generalized cloning channels

For qudits, Schur–Weyl duality gives

$$V_{\text{Schur}} : (\mathbb{C}^d)^{\otimes n} \rightarrow \bigoplus_{\lambda} \mathcal{H}_{\lambda} \otimes \mathcal{M}_{\lambda} \quad (11)$$

where $\lambda \vdash n$ is a partition with $\lambda_1 \geq \dots \geq \lambda_d \geq 0$, $\mathcal{H}_{\lambda}$ is the corresponding $GL(d, \mathbb{C})$ irrep of dimension d_{λ} , and $\mathcal{M}_{\lambda}$ is the symmetric-group irrep.

For a density matrix ρ on $\mathbb{C}^d$, the Schur decomposition is

$$V_{\text{Schur}} \rho^{\otimes n} V_{\text{Schur}}^\dagger = \bigoplus_{\lambda \vdash n} q_{\lambda, n} \rho_{\lambda} \otimes \tau_{\lambda}, \quad (12)$$

where ρ_{λ} is the normalized $GL(d, \mathbb{C})$ irrep state of ρ , $\tau_{\lambda} := \mathbb{I}_{\mathcal{M}_{\lambda}} / \dim \mathcal{M}_{\lambda}$ is the maximally mixed state on $\mathcal{M}_{\lambda}$, and

$$q_{\lambda, n} := s_{\lambda}(x) \dim \mathcal{M}_{\lambda}, \quad (13)$$

with x the spectrum of ρ and s_{λ} the Schur polynomial.

Following YCH, we measure the Schur block label and obtain some outcome λ . The key is to find a $U(d)$ -covariant channel that maps every resulting irrep state to one fixed target irrep; Section 4 chooses a padded target that dominates the typical set.

¹The cloning map is usually written as $\mathcal{C}_{n \rightarrow m}(\rho) = \frac{d_{\text{Sym}^n(\mathbb{C}^d)}}{d_{\text{Sym}^m(\mathbb{C}^d)}} P_{\text{Sym}^m}(\rho \otimes \mathbb{I}_{m-n}) P_{\text{Sym}^m}$, where P_{Sym^m} projects onto $\text{Sym}^m(\mathbb{C}^d)$ inside $\text{Sym}^n(\mathbb{C}^d) \otimes \text{Sym}^{m-n}(\mathbb{C}^d)$. We use the intrinsic isometry V so that the codomain is $\text{Sym}^m(\mathbb{C}^d)$.

$$\begin{array}{ccc}
\mathcal{B}(\mathcal{H}_\mu) & \xrightarrow{\mathcal{N}} & \mathcal{B}(\mathcal{H}_\nu) \\
\mathcal{U}_\mu(g) \downarrow & & \downarrow \mathcal{U}_\nu(g) \\
\mathcal{B}(\mathcal{H}_\mu) & \xrightarrow{\mathcal{N}} & \mathcal{B}(\mathcal{H}_\nu)
\end{array}$$

Figure 1: Commutative diagram for the covariance condition (14): applying $\mathcal{U}_\mu(g)$ before the channel equals applying $\mathcal{U}_\nu(g)$ after it.

3.1 Generalized cloner from the PRV component

We use finite-dimensional rational irreducible representations of $\mathrm{GL}(d, \mathbb{C})$, restricted to $\mathrm{U}(d)$ when discussing covariance. Their highest weights are dominant integral vectors and may have negative entries. Partitions label the polynomial representations, which extend to all matrices in $M_d(\mathbb{C})$. The input and output weights in the state-compression theorem are partitions; auxiliary and dual representations may have general dominant integral highest weights.

More explicitly, a highest weight λ gives $\pi_\lambda = \det^{\lambda_d} \otimes \pi_{\lambda - \lambda_d(1, \dots, 1)}$. Thus $\pi_{(1, -1)} = \det^{-1} \otimes \pi_{(2, 0)}$ for $\mathrm{GL}(2)$. A determinant twist cancels from the unitary conjugation action and from the normalized representation state of an invertible density matrix. It need not extend to singular matrices: negative determinant powers require invertibility. We therefore retain general dominant auxiliary weights and state the domains explicitly.

A channel $\mathcal{N} : \mathcal{B}(\mathcal{H}_\mu) \rightarrow \mathcal{B}(\mathcal{H}_\nu)$ between irreducible representation spaces is $\mathrm{U}(d)$ -covariant if

$$\mathcal{N} \circ \mathcal{U}_\mu(g) = \mathcal{U}_\nu(g) \circ \mathcal{N}, \quad \forall g \in \mathrm{U}(d), \quad (14)$$

where $\mathcal{U}_\mu(g)(X) := U_\mu(g) X U_\mu(g)^\dagger$, and similarly for ν ; see Figure 1.

Equivalently, the Choi operator satisfies [5, Lemma 11]

$$[J_{\mathcal{N}}, U_\mu^* \otimes U_\nu] = 0. \quad (15)$$

The tensor product of the dual representation μ^* with ν decomposes as

$$\mathcal{H}_{\mu^*} \otimes \mathcal{H}_\nu \simeq \bigoplus_{\lambda} \mathcal{H}_\lambda \otimes \mathbb{C}^{m_\lambda}. \quad (16)$$

Schur's lemma therefore gives the general commutant form

$$J_{\mathcal{N}} \simeq \bigoplus_{\lambda} \mathbb{I}_{\mathcal{H}_\lambda} \otimes A_\lambda, \quad A_\lambda \geq 0, \quad (17)$$

with the trace-preserving condition imposing additional linear constraints on the matrices A_λ . In a multiplicity-one sector, the normalized projector $(d_\mu/d_\lambda)\Pi_\lambda$ is itself the Choi operator of a covariant channel: its partial trace over $\mathcal{H}_\nu$ is $\mathbb{I}_{\mu^*}$ by Schur's lemma and trace normalization. The construction below uses precisely such a sector. The structure in (17) may be viewed as a simpler, irrep-level version of the more general covariant-channel classification in [24].

We choose the projector onto the PRV component, denoted by ω . Its Cartan intertwiner formula and the approximation bounds in Theorem 2 provide the properties needed for compression. The representation is minimal in the dominance order, which for equal total weights $|\lambda| := \sum_i \lambda_i$ is defined by

$$\lambda \preceq \eta \iff |\lambda| = |\eta| \text{ and } \sum_{i=1}^k \lambda_i \leq \sum_{i=1}^k \eta_i \quad (1 \leq k < d). \quad (18)$$

The tensor product has a unique minimum in this order, occurring with multiplicity one: the *Parthasarathy–Ranga Rao–Varadarajan (PRV) component* [25, 26]. Since the dual highest weight is $\mu^* = -w_0(\mu)$, it is

$$\omega = (\mu^* + w_0(\nu))^+ = (\nu - \mu)^+. \quad (19)$$

Here w_0 reverses coordinates, and $(\lambda)^+$ sorts them into dominant order $\lambda_1 \geq \dots \geq \lambda_d$. In the Cartan case, $\nu - \mu$ is already dominant and $\omega = \nu - \mu$.

For the state comparisons in this Cartan case, μ and ν are partitions with at most $r = \text{rank } \rho$ rows. When $r = d$, the auxiliary highest weight ω may have negative entries; its finite-dimensional rational representation can be evaluated at ρ because ρ is invertible. When $r < d$, we have $\omega_{r+1} = \dots = \omega_d = 0$, so dominance forces ω to be a partition.

We therefore define our *generalized cloning map* as follows.

Definition 1 (Generalized Cloning Map). *Let μ and ν be $\text{GL}(d, \mathbb{C})$ irreps, and let $\omega = (\nu - \mu)^+$ be the PRV component in the decomposition of $\mu^* \otimes \nu$. The generalized covariant cloning map $\mathcal{C}_{\mu \rightarrow \nu} : \mathcal{B}(\mathcal{H}_\mu) \rightarrow \mathcal{B}(\mathcal{H}_\nu)$ is defined as*

$$\mathcal{C}_{\mu \rightarrow \nu}(\sigma_\mu) := \text{Tr}_\mu \left[J_\omega(\sigma_\mu^\top \otimes \mathbb{I}_\nu) \right], \quad J_\omega := \frac{d_\mu}{d_\omega} \Pi_\omega. \quad (20)$$

Multiplicity one makes Π_ω unambiguous. The compression analysis uses the Cartan case on typical pairs; this general definition also supplies channels for atypical pairs.

The map $\mathcal{C}_{\mu \rightarrow \nu}$ is completely positive because J_ω is proportional to a projector. Covariance, Schur's lemma, and trace normalization give $\text{Tr}_\nu(\Pi_\omega) = (d_\omega/d_\mu)\mathbb{I}_{\mu^*}$. Hence

$$\text{Tr}_\nu(J_\omega) = \frac{d_\mu}{d_\omega} \text{Tr}_\nu(\Pi_\omega) = \mathbb{I}_{\mu^*},$$

so the map is trace-preserving.

For symmetric powers this recovers Werner's cloner. For general irreps, the map acts on representation states; the next formula explains its cloning interpretation.

3.2 Properties of the generalized cloner

3.2.1 Cartan intertwiner formula

In the Cartan case $\nu = \mu + \omega$, an intertwining isometry gives a representation that makes the connection to Werner's cloning map explicit (6).

Proposition 1 (Cartan intertwiner formula). *Let μ, ν, ω be dominant integral highest weights with $\nu = \mu + \omega$, and let $V : \mathcal{H}_\nu \rightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega$ be the isometric embedding into the Cartan component, characterized up to phase by*

$$VU_\nu(g) = (U_\mu(g) \otimes U_\omega(g))V \quad \text{for all } g \in \text{U}(d). \quad (21)$$

Then the covariant channel

$$\mathcal{C}_{\mu \rightarrow \nu}(\sigma_\mu) = \frac{d_\mu}{d_\nu} V^\dagger(\sigma_\mu \otimes \mathbb{I}_\omega)V \quad (22)$$

has Choi operator $J = (d_\mu/d_\omega)\Pi_\omega$, where Π_ω projects onto the multiplicity-one PRV component $(\nu - \mu)^+$ in $\mu^ \otimes \nu$.*

Proof. See Appendix A. □

To interpret the channel, we factor the target state through the same embedding and identify the auxiliary state replaced by the maximally mixed state.

Complexifying the Lie-algebra intertwining relation in Proposition 1 extends it from $U(d)$ to $GL(d, \mathbb{C})$:

$$V\pi_\nu(A) = (\pi_\mu(A) \otimes \pi_\omega(A))V \quad A \in GL(d, \mathbb{C}) .$$

When all three highest weights are partitions, the representation matrices are polynomial and this identity extends to all $A \in M_d(\mathbb{C})$.

For the state comparison between $\mathcal{C}_{\mu \rightarrow \nu}(\rho_\mu)$ and ρ_ν , take μ, ν to be partitions of length at most $r = \text{rank } \rho$ and $\omega = \nu - \mu$ to be dominant. Define $s_\omega(x) := \text{Tr} \pi_\omega(\rho)$ and $\rho_\omega := \pi_\omega(\rho)/s_\omega(x)$; s_ω is a Schur polynomial when ω is a partition and a Schur Laurent polynomial otherwise. Taking $A = \rho$ and using $V^\dagger V = \mathbb{I}_\nu$ gives the exact factorization

$$\rho_\nu = \frac{s_\mu(x) s_\omega(x)}{s_\nu(x)} V^\dagger (\rho_\mu \otimes \rho_\omega) V , \quad (23)$$

to be compared with

$$\mathcal{C}_{\mu \rightarrow \nu}(\rho_\mu) = \frac{d_\mu d_\omega}{d_\nu} V^\dagger \left(\rho_\mu \otimes \frac{\mathbb{I}_\omega}{d_\omega} \right) V . \quad (24)$$

The factor ρ_ω depends on the same unknown state as ρ_μ . The channel replaces it by the maximally mixed state, the unique $U(d)$ -invariant state on $\mathcal{H}_\omega$, and projects onto the Cartan component. For symmetric powers $\mu = (n)$ and $\nu = (m)$ with $m \geq n$, this is Werner's construction: $\rho_{(m)} \propto V^\dagger (\rho_{(n)} \otimes \rho_{(m-n)}) V$, with the auxiliary representation state replaced by the identity (6). For pure $\rho = |\psi\rangle\langle\psi|$ only symmetric powers carry weight and $\rho_{(n)} = |\psi\rangle\langle\psi|^{\otimes n}$, giving literal cloning of copies. Theorem 2 bounds the resulting error in trace distance.

Beyond the resemblance to Werner's cloning map, one of us has recently used the generalized cloning map (for $\nu = \mu + \omega$) as a key construct in a protocol for mixed-states cloning [27], which is shown to be asymptotically optimal.

Furthermore, Werner's cloner has also been applied in the quantum purity amplification for qubits [28]. It turns out a special case of generalized cloning map also arises in qudit quantum purity amplification [29, 30, 31], the (asymptotically) optimal protocol is the generalized cloning map from μ to the fundamental irrep $\square$, and its Choi matrix projects onto the PRV component of $\mu^* \otimes \square$.

Hence, this generalization we propose is both formal and operational.

3.2.2 Identity and qubit cases

When $\mu = \nu$, the PRV component is trivial and the channel is the identity.

For $GL(2)$, the irrep with highest weight (a, b) is $\det^b \otimes \text{Sym}^{a-b}(\mathbb{C}^2)$. The determinant factor cancels from the conjugation action, so the comparison of covariant channels reduces to symmetric powers.

Remark 1. Let $\mathcal{H}_\mu = \text{Sym}^n(\mathbb{C}^2)$ and $\mathcal{H}_\nu = \text{Sym}^m(\mathbb{C}^2)$. For $m \geq n$, the generalized cloning map reduces to Werner's cloning map (6); for $m < n$, it is the partial-trace channel.

Dualizing the input gives $\mu^* \simeq \det^{-n} \otimes \text{Sym}^n(\mathbb{C}^2)$. The Clebsch–Gordan decomposition therefore reads:

$$\mu^* \otimes \nu \simeq \bigoplus_{k=0}^{\min(m,n)} \det^{k-n} \otimes \text{Sym}^{m+n-2k}(\mathbb{C}^2) . \quad (25)$$

The PRV component is the term $k = \min(m, n)$, with highest weight $(m - n, 0)$ for $m \geq n$ and $(0, m - n)$ for $m < n$. Its dimension, and hence the channel's Kraus rank, is $|m - n| + 1$. Upon restriction to $\text{SU}(2)$, the decomposition can equivalently be written $\bigoplus_{k=0}^{\min(m, n)} \text{Sym}^{|m-n|+2k}(\mathbb{C}^2)$. These channels are exactly those used by YCH in (8).

3.2.3 Dominance order and dimension

Remark 2 (Minimal $\neq$ Smallest). *The PRV component in $\mu^* \otimes \nu$ is not, in general, the irreducible summand of smallest dimension. This failure can occur even in the dominant case $\nu = \mu + \omega$. For example, let $\mu = (1, 0, 0)$ and $\nu = (5, 1, 0)$ be $\text{GL}(3)$ irreps. The PRV component is $\omega := (\nu - \mu)^+ = \nu - \mu = (4, 1, 0)$, which has dimension 24. However, $(5, 0, 0)$ is also a summand in the decomposition of $\mu^* \otimes \nu^2$ and it is the smallest summand of dimension 21.*

3.2.4 Reverse cloner

Reversing the representation pair gives the Petz recovery map with respect to the maximally mixed input.

Proposition 2 (Reverse cloner). *Let $\mathcal{C}_{\mu \rightarrow \nu} : \mathcal{B}(\mathcal{H}_\mu) \rightarrow \mathcal{B}(\mathcal{H}_\nu)$ be the generalized cloning map, and let $\sigma_\mu := \mathbb{I}_\mu / d_\mu$. Then the Petz recovery map of $\mathcal{C}_{\mu \rightarrow \nu}$ with respect to σ_μ is exactly the reverse cloning map:*

$$\mathcal{R}_{\sigma_\mu, \mathcal{C}_{\mu \rightarrow \nu}} = \mathcal{C}_{\nu \rightarrow \mu}.$$

Equivalently,

$$\mathcal{C}_{\nu \rightarrow \mu}(X) = \frac{d_\nu}{d_\mu} \mathcal{C}_{\mu \rightarrow \nu}^\dagger(X),$$

where $\mathcal{C}_{\mu \rightarrow \nu}^\dagger$ is the adjoint map.

Proof. See Appendix A. □

In the pure state case, the reverse cloner corresponds to tracing out the excess systems.

3.2.5 Approximation of representation states

We compare $\mathcal{C}_{\mu \rightarrow \nu}(\rho_\mu)$ with ρ_ν , and similarly in the reverse direction. The error is controlled by the size of the dominant difference relative to the row gaps of μ .

Theorem 2 (Cloning accuracy). *Let ρ be a density matrix on $\mathbb{C}^d$, $d \geq 2$, of rank $r \leq d$ with strictly decreasing positive eigenvalues $x_1 > \dots > x_r > 0$. Let μ, ν be partitions with at most r rows, and suppose that $\omega := \nu - \mu$ is a dominant integral weight. Set*

$$D := \|\omega\|_1, \quad b_\mu := \min_{1 \leq i \leq \min(r, d-1)} (\mu_i - \mu_{i+1}).$$

For the normalized irrep states ρ_μ, ρ_ν , the generalized cloning channels satisfy, for a constant $C_{d,x}$ depending only on the dimension and the nonzero spectrum,

$$\begin{aligned} \frac{1}{2} \|\mathcal{C}_{\mu \rightarrow \nu}(\rho_\mu) - \rho_\nu\|_1 &\leq C_{d,x} \frac{D}{b_\mu + 1}, \\ \frac{1}{2} \|\mathcal{C}_{\nu \rightarrow \mu}(\rho_\nu) - \rho_\mu\|_1 &\leq C_{d,x} \frac{D}{b_\mu + 1}. \end{aligned} \tag{26}$$

²This is because $(1, 0, 0) \otimes (5, 0, 0) \cong (6, 0, 0) \oplus (5, 1, 0)$ from the Pieri rule.

The case $d = 1$ is trivial, since both representation spaces are one-dimensional. For $r < d$, the gap b_μ includes $\mu_r - \mu_{r+1} = \mu_r$; for $r = d$, it involves only adjacent row differences. In particular, $b_\mu = \Omega(n)$ gives errors $O(D/n)$, which vanish if $D = o(n)$. This trace-distance estimate is used in the achievability analysis in the following section and will be proven in Section 7.

Covariance also gives a useful structural property of the output.

Proposition 3 (Commutativity). *Let ρ have rank r , let μ, ν be partitions with at most r rows, and let $\mathcal{N} : \mathcal{B}(\mathcal{H}_\mu) \rightarrow \mathcal{B}(\mathcal{H}_\nu)$ be any $U(d)$ -covariant channel. Then*

$$[\mathcal{N}(\rho_\mu), \rho_\nu] = 0.$$

Proof. In an eigenbasis of ρ , the state ρ_μ is invariant under the diagonal unitary torus. By covariance, so is $\mathcal{N}(\rho_\mu)$, which is therefore block diagonal in the weight spaces of $\mathcal{H}_\nu$. Since ρ_ν is scalar on each such space, the two operators commute. $\square$

4 Achievability

The qudit protocol maps all Schur blocks into one padded target representation. For a rank- r source with $x_1 > \dots > x_r > 0$, define the typical set by

$$\epsilon_n := \sqrt{n/2 \log n}, \quad \mathcal{T}_{x,n} := \left\{ \lambda \vdash n : \ell(\lambda) \leq r, \max_{1 \leq i \leq r} |\lambda_i - nx_i| \leq \epsilon_n \right\}. \quad (27)$$

Define the fixed target representation by

$$\Lambda_i := \begin{cases} \lceil nx_i + (r - i + 1)(2\epsilon_n + 1) \rceil, & 1 \leq i \leq r, \\ 0, & r < i \leq d. \end{cases} \quad (28)$$

Thus Λ is a partition of an integer slightly larger than n . The compression scheme is as follows.

Qudit Compression Scheme

Encoding (34):

1. **Irrep Measurement:** Measure the Schur label λ without disturbing the state within $\mathcal{H}_\lambda \otimes \mathcal{M}_\lambda$.
2. **Cloning:** Discard the multiplicity register $\mathcal{M}_\lambda$ and apply $\mathcal{C}_{\lambda \rightarrow \Lambda}$, with the target in (28), to the remaining state ρ_λ . Store the output in a memory of dimension $\dim \mathcal{H}_\Lambda$.

Decoding (36):

1. **Irrep Sampling:** Pick a target representation λ' at random from the known Schur–Weyl distribution $q_{\lambda',n}$.
2. **Cloning:** Apply the generalized cloning channel $\mathcal{C}_{\Lambda \rightarrow \lambda'}$ to the state in the quantum memory.
3. **State Reconstruction:** Append a new multiplicity register initialized in the maximally mixed state $\tau_{\lambda'}$, and perform the inverse Schur transform.

This protocol achieves the memory cost in Theorem 1.

Theorem 3 (Achievability). *Let ρ be a density matrix with rank $r \leq d$ and strictly decreasing non-zero spectrum $x_1 > \dots > x_r > 0$. Let $\rho_g := U(g)\rho U(g)^\dagger$ for $g \in U(d)$. There exists a sequence of compression codes $(\mathcal{E}_n, \mathcal{D}_n)$ with quantum memory M_n of size*

$$|M_n| = L_{d,r}(n, x) + o(1),$$

where $L_{d,r}(n, x)$ is defined in (2), and error

$$\delta_n := \sup_{g \in U(d)} \frac{1}{2} \left\| \mathcal{D}_n \circ \mathcal{E}_n(\rho_g^{\otimes n}) - \rho_g^{\otimes n} \right\|_1 = O\left(\frac{\log n}{\sqrt{n}}\right).$$

The implicit constant depends only on d, r and the fixed spectrum x .

For pure states, we have

$$|M_n| = (d-1) \log n - \log(d-1)! + o(1). \quad (29)$$

The memory size is $\log \dim \mathcal{H}_\Lambda$, which we evaluate using the following consequence of the Weyl dimension formula.

Lemma 1 (Asymptotic Weyl dimension). *Fix d and a spectrum with $x_1 > \dots > x_r > 0$ and $x_i = 0$ for $i > r$. Let λ be a partition, not necessarily of n , with $\lambda_i = nx_i + O(\sqrt{n}\zeta_n)$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$, where $\zeta_n = o(\sqrt{n})$. Then*

$$\log \dim \mathcal{H}_\lambda = L_{d,r}(n, x) + O\left(\frac{\zeta_n}{\sqrt{n}} + \frac{1}{n}\right). \quad (30)$$

Proof. The exact dimension is given by the Weyl dimension formula:

$$\dim \mathcal{H}_\lambda = \prod_{1 \leq i < j \leq d} \frac{\lambda_i - \lambda_j + j - i}{j - i}. \quad (31)$$

For $i < j \leq r$ the numerator is $n(x_i - x_j)[1 + O(\zeta_n/\sqrt{n} + 1/n)]$, while for $i \leq r < j$ it is $nx_i[1 + O(\zeta_n/\sqrt{n} + 1/n)]$. The factors with $r < i < j$ equal $j - i$ and cancel their denominators exactly. Since d is fixed, multiplying the remaining factors gives

$$\dim \mathcal{H}_\lambda = \frac{n^{\frac{r(2d-r-1)}{2}} \prod_{1 \leq i < j \leq r} (x_i - x_j) \prod_{i=1}^r x_i^{d-r}}{\prod_{k=d-r}^{d-1} k!} \left[1 + O\left(\frac{\zeta_n}{\sqrt{n}} + \frac{1}{n}\right) \right], \quad (32)$$

where the factorial ratio follows from $\prod_{i < j} (j - i) = \prod_{k=1}^{d-1} k!$ (with the usual empty-product convention). Taking the logarithm proves the claim. $\square$

Proof of Theorem 3. For $d = 1$, the source is a single known state and no memory is needed. Assume henceforth that $d \geq 2$. We omit the subscript g in ρ_g in this proof. Every channel in the protocol is $U(d)$ -covariant and the weights $q_{\lambda,n}$ are independent of g , so the pointwise error is the same throughout the orbit. By Schur–Weyl duality, since ρ has rank r , the state $\rho^{\otimes n}$ is supported on irreps with at most r rows. Thus it is unitarily equivalent to the block-diagonal state

$$\rho^{\otimes n} \simeq \bigoplus_{\lambda: \ell(\lambda) \leq r} q_{\lambda,n} (\rho_\lambda \otimes \tau_\lambda). \quad (33)$$

Let P_λ be the projection onto the block labeled by λ , and set $P := \sum_{\lambda: \ell(\lambda) \leq r} P_\lambda$, so every source state is supported in P . The encoder is the channel on all of $(\mathbb{C}^d)^{\otimes n}$ defined by

$$\mathcal{E}_n(X) = \sum_{\lambda: \ell(\lambda) \leq r} \mathcal{C}_{\lambda \rightarrow \Lambda}(\text{Tr}_{\mathcal{M}_\lambda}(P_\lambda X P_\lambda)) + \text{Tr}[(\mathbb{I} - P)X] \frac{\mathbb{I}_{\mathcal{H}_\Lambda}}{\dim \mathcal{H}_\Lambda}, \quad (34)$$

where the complement term makes $\mathcal{E}_n$ completely positive and trace preserving on the full input space. It vanishes on every source state $\rho_g^{\otimes n}$, whose support lies in P . Here $\mathcal{C}_{\lambda \rightarrow \Lambda}$ is defined by (20), with target (28).

This padding ensures that $\Lambda - \lambda$ is dominant for every $\lambda \in \mathcal{T}_{x,n}$. Indeed, typicality and $[a] - [b] \geq a - b - 1$ give, for $1 \leq i < r$,

$$\Lambda_i - \Lambda_{i+1} \geq n(x_i - x_{i+1}) + 2\epsilon_n \geq \lambda_i - \lambda_{i+1}. \quad (35)$$

Moreover, $(\Lambda - \lambda)_r \geq \epsilon_n + 1 > 0$. Thus the rows of $\Lambda - \lambda$ are non-increasing and non-negative.

The corresponding decoder maps back to the original blocks:

$$\mathcal{D}_n(\cdot) = \bigoplus_{\lambda: \ell(\lambda) \leq r} q_{\lambda,n} (\mathcal{C}_{\Lambda \rightarrow \lambda}(\cdot) \otimes \tau_\lambda). \quad (36)$$

The decoder samples λ' independently of the measured λ , so the composition contains cross terms with $\lambda \neq \lambda'$. We bound its error through the reference state ρ_Λ . The encoded memory state is

$$\mathcal{E}_n(\rho^{\otimes n}) = \sum_{\lambda: \ell(\lambda) \leq r} q_{\lambda,n} \mathcal{C}_{\lambda \rightarrow \Lambda}(\rho_\lambda). \quad (37)$$

By the triangle inequality and the contractivity of the trace distance under $\mathcal{D}_n$,

$$\begin{aligned} \delta_n &\leq \frac{1}{2} \|\mathcal{D}_n \circ \mathcal{E}_n(\rho^{\otimes n}) - \mathcal{D}_n(\rho_\Lambda)\|_1 + \frac{1}{2} \|\mathcal{D}_n(\rho_\Lambda) - \rho^{\otimes n}\|_1 \\ &\leq \frac{1}{2} \|\mathcal{E}_n(\rho^{\otimes n}) - \rho_\Lambda\|_1 + \sum_{\lambda'} q_{\lambda',n} \frac{1}{2} \|\mathcal{C}_{\Lambda \rightarrow \lambda'}(\rho_\Lambda) - \rho_{\lambda'}\|_1, \end{aligned} \quad (38)$$

where we used the block-diagonal identity

$$\mathcal{D}_n(\rho_\Lambda) - \rho^{\otimes n} = \bigoplus_{\lambda'} q_{\lambda',n} (\mathcal{C}_{\Lambda \rightarrow \lambda'}(\rho_\Lambda) - \rho_{\lambda'}) \otimes \tau_{\lambda'}$$

and the multiplicativity of the trace norm under tensoring with $\tau_{\lambda'}$. Bounding the first term by convexity of the trace norm,

$$\frac{1}{2} \|\mathcal{E}_n(\rho^{\otimes n}) - \rho_\Lambda\|_1 \leq \sum_{\lambda} q_{\lambda,n} \frac{1}{2} \|\mathcal{C}_{\lambda \rightarrow \Lambda}(\rho_\lambda) - \rho_\Lambda\|_1, \quad (39)$$

we arrive at

$$\delta_n \leq \sum_{\lambda} q_{\lambda,n} \frac{1}{2} \|\mathcal{C}_{\lambda \rightarrow \Lambda}(\rho_\lambda) - \rho_\Lambda\|_1 + \sum_{\lambda'} q_{\lambda',n} \frac{1}{2} \|\mathcal{C}_{\Lambda \rightarrow \lambda'}(\rho_\Lambda) - \rho_{\lambda'}\|_1. \quad (40)$$

Each of the two sums in (40) splits into typical and atypical contributions:

$$\delta_n \leq \max_{\lambda \in \mathcal{T}_{x,n}} \frac{1}{2} \|\mathcal{C}_{\lambda \rightarrow \Lambda}(\rho_\lambda) - \rho_\Lambda\|_1 + \max_{\lambda' \in \mathcal{T}_{x,n}} \frac{1}{2} \|\mathcal{C}_{\Lambda \rightarrow \lambda'}(\rho_\Lambda) - \rho_{\lambda'}\|_1 + 2 \sum_{\lambda \notin \mathcal{T}_{x,n}} q_{\lambda,n}, \quad (41)$$

having used the maximal trace distance bound $\frac{1}{2} \|\rho - \sigma\|_1 \leq 1$ for the atypical terms in each sum; this is where the factor of 2 on the tail term comes from.

The tail mass is controlled by Sanov's theorem via the following lemma:

Lemma 2 (Sanov's theorem [20, 32, 33, 34]). *Let λ/n be the empirical probability distribution associated with a Young diagram distributed according to $q_{\lambda,n}$. Define the total variation distance by*

$$d(\lambda/n, x) := \frac{1}{2} \sum_{i=1}^d |\lambda_i/n - x_i|.$$

Then, for every $t > 0$,

$$\Pr[\lambda : d(\lambda/n, x) > t] \leq (n+1)^{r(r+1)/2} e^{-2nt^2}.$$

Lemma 2 follows from the pointwise bound $\Pr[\lambda] \leq (n+1)^{r(r-1)/2} 2^{-nD(\lambda/n\|x)}$ on the supported Schur–Weyl diagrams (see [32]). Summing over at most $(n+1)^r$ diagrams and using Pinsker's inequality $D(\lambda/n\|x) \geq (2/\ln 2) d(\lambda/n, x)^2$ gives the claimed event bound. The distribution $q_{\lambda,n}$ is supported on diagrams with at most r rows. Hence, if λ lies in its support but not in $\mathcal{T}_{x,n}$, then for some i , $|\lambda_i/n - x_i| > (\log n)/\sqrt{2n}$, and hence $d(\lambda/n, x) > (\log n)/(2\sqrt{2n})$. Lemma 2 therefore gives

$$\sum_{\lambda \notin \mathcal{T}_{x,n}} q_{\lambda,n} \leq (n+1)^{r(r+1)/2} e^{-(\log n)^2/4}, \quad (42)$$

which is superpolynomially small.

For typical states $\lambda \in \mathcal{T}_{x,n}$, the two maxima in (41) are the forward and reverse cloning errors of Theorem 2, applied with $\mu = \lambda$ and $\nu = \Lambda$. The dominance hypothesis holds by construction, and $b_\lambda = \Omega(n)$ uniformly on the typical set because the positive eigenvalues are distinct and, when $r < d$, $x_r > 0$.

Since $\Lambda - \lambda$ is non-negative and $|\lambda| = n$, its distance from every typical λ is

$$\|\Lambda - \lambda\|_1 = \sum_i (\Lambda_i - \lambda_i) = |\Lambda| - n = O(\epsilon_n) = O(\sqrt{n} \log n). \quad (43)$$

Theorem 2 gives the trace-distance estimate directly, uniformly over typical λ , in both directions:

$$\max \left\{ \frac{1}{2} \|\mathcal{C}_{\lambda \rightarrow \Lambda}(\rho_\lambda) - \rho_\Lambda\|_1, \frac{1}{2} \|\mathcal{C}_{\Lambda \rightarrow \lambda}(\rho_\Lambda) - \rho_\lambda\|_1 \right\} = O\left(\frac{\log n}{\sqrt{n}}\right). \quad (44)$$

Thus each typical maximum in (41) has this order, and (42) yields

$$\delta_n \leq O\left(\frac{\log n}{\sqrt{n}}\right) + 2(n+1)^{\frac{r(r+1)}{2}} e^{-(\log n)^2/4} = O\left(\frac{\log n}{\sqrt{n}}\right). \quad (45)$$

The memory is $\mathcal{H}_\Lambda$. Since $\Lambda_i = nx_i + O(\sqrt{n} \log n)$ for $i \leq r$ and $\Lambda_i = 0$ for $i > r$, Lemma 1 with $\zeta_n = \log n$ gives

$$|M_n| = \log \dim \mathcal{H}_\Lambda = L_{d,r}(n, x) + o(1).$$

□

5 Converse

The Koashi–Imoto theorem characterizes exact compression [15, 16]. For approximate recovery, we prove a memory bound for irreducible orbits under a spectral-gap assumption. It applies to arbitrary encoding and decoding channels.

Proposition 4 (Memory bound for irreducible orbits). *Let $U : G \rightarrow \mathcal{U}(\mathcal{H})$ be a continuous irreducible unitary representation of a compact group on a finite-dimensional space $\mathcal{H}$, with normalized Haar measure dg . Let τ be a state on $\mathcal{H}$ with a simple largest eigenvalue p_0 and corresponding rank-one projector P . Set*

$$p_1 := \|(\mathbb{I} - P)\tau(\mathbb{I} - P)\|_\infty, \quad \gamma := p_0 - p_1 > 0, \quad \tau_g := U(g)\tau U(g)^\dagger.$$

For any finite-dimensional memory M and encoding and decoding channels $\mathcal{E} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(M)$ and $\mathcal{D} : \mathcal{B}(M) \rightarrow \mathcal{B}(\mathcal{H})$, define

$$\bar{\delta} := \int_G \frac{1}{2} \|\mathcal{D} \circ \mathcal{E}(\tau_g) - \tau_g\|_1 dg.$$

Then

$$\dim M \geq \dim \mathcal{H} \left(1 - \frac{\bar{\delta}}{\gamma}\right). \quad (46)$$

Proof. Write $\Phi = \mathcal{D} \circ \mathcal{E}$ and $P_g = U(g)PU(g)^\dagger$. Trace preservation makes $\mathcal{E}(P_g)$ a state, so $\mathcal{E}(P_g) \leq \mathbb{I}_M$. Combining this with $\tau_g \leq p_1 \mathbb{I} + \gamma P_g$, positivity gives

$$\Phi(\tau_g) \leq p_1 \Phi(\mathbb{I}) + \gamma \mathcal{D}(\mathbb{I}_M).$$

Schur's lemma gives $\int_G P_g dg = \mathbb{I} / \dim \mathcal{H}$. Testing against P_g , averaging, and using trace preservation therefore yields

$$p_0 - \bar{\delta} \leq \int_G \text{Tr}[P_g \Phi(\tau_g)] dg \leq \frac{p_1 \text{Tr} \Phi(\mathbb{I}) + \gamma \text{Tr} \mathcal{D}(\mathbb{I}_M)}{\dim \mathcal{H}} = p_1 + \gamma \frac{\dim M}{\dim \mathcal{H}}.$$

Rearranging proves the claim. $\square$

At zero error this recovers the Koashi–Imoto incompressibility bound for these orbit families: $\dim M \geq \dim \mathcal{H}$. For the representation states used here, the spectral gap is bounded below uniformly in the partition.

Lemma 3 (Uniform spectral gap). *Let ρ have strictly decreasing nonzero spectrum $x_1 > \dots > x_r > 0$. Set*

$$q_x := \max_{1 \leq i < r} \frac{x_{i+1}}{x_i}, \quad \gamma_x := (1 - q_x) \prod_{1 \leq i < j \leq r} \left(1 - \frac{x_j}{x_i}\right) > 0,$$

with $q_x := 0$ for $r = 1$. For every partition λ with at most r rows, ρ_λ has a simple largest eigenvalue and its gap γ , defined as in Proposition 4, satisfies $\gamma \geq \gamma_x$.

Proof. See Appendix A. $\square$

We transfer an arbitrary compression code to the padded target representation and apply these two bounds. Haar-average error is sufficient, so the converse result also applies to the worst-case error criterion (1).

Theorem 4 (Converse). *Let ρ have rank $r \leq d$ and strictly decreasing nonzero spectrum $x_1 > \dots > x_r > 0$, and write $\rho_g = U(g)\rho U(g)^\dagger$. Suppose a sequence of compression codes $(\mathcal{E}_n, \mathcal{D}_n)$ with memory M_n has Haar-average error*

$$\bar{\delta}_n := \int_{U(d)} \frac{1}{2} \|\mathcal{D}_n \circ \mathcal{E}_n(\rho_g^{\otimes n}) - \rho_g^{\otimes n}\|_1 dg \longrightarrow 0.$$

Then

$$|M_n| \geq L_{d,r}(n, x) + o(1),$$

where $L_{d,r}(n, x)$ is defined in (2).

Proof. Let $\mathcal{A}_n$ and $\mathcal{B}_n$ be the achievability encoder and decoder in (34) and (36). Write $\tau_{g,n} := U_\Lambda(g)\rho_\Lambda U_\Lambda(g)^\dagger$. The forward and reverse estimates in the achievability proof give

$$e_n := \sup_g \left[\frac{1}{2} \|\mathcal{A}_n(\rho_g^{\otimes n}) - \tau_{g,n}\|_1 + \frac{1}{2} \|\mathcal{B}_n(\tau_{g,n}) - \rho_g^{\otimes n}\|_1 \right] = O\left(\frac{\log n}{\sqrt{n}}\right).$$

The channels $\mathcal{E}_n \circ \mathcal{B}_n$ and $\mathcal{A}_n \circ \mathcal{D}_n$ form a code for the target orbit using the same memory M_n . For each g , insert the intermediate states $\mathcal{A}_n \circ \mathcal{D}_n \circ \mathcal{E}_n(\rho_g^{\otimes n})$ and $\mathcal{A}_n(\rho_g^{\otimes n})$. The triangle inequality gives

$$\begin{aligned} & \frac{1}{2} \|\mathcal{A}_n \circ \mathcal{D}_n \circ \mathcal{E}_n \circ \mathcal{B}_n(\tau_{g,n}) - \tau_{g,n}\|_1 \\ & \leq \frac{1}{2} \|\mathcal{A}_n \circ \mathcal{D}_n \circ \mathcal{E}_n(\mathcal{B}_n(\tau_{g,n}) - \rho_g^{\otimes n})\|_1 + \frac{1}{2} \|\mathcal{A}_n(\mathcal{D}_n \circ \mathcal{E}_n(\rho_g^{\otimes n}) - \rho_g^{\otimes n})\|_1 + \frac{1}{2} \|\mathcal{A}_n(\rho_g^{\otimes n}) - \tau_{g,n}\|_1 \\ & \leq \frac{1}{2} \|\mathcal{B}_n(\tau_{g,n}) - \rho_g^{\otimes n}\|_1 + \frac{1}{2} \|\mathcal{D}_n \circ \mathcal{E}_n(\rho_g^{\otimes n}) - \rho_g^{\otimes n}\|_1 + \frac{1}{2} \|\mathcal{A}_n(\rho_g^{\otimes n}) - \tau_{g,n}\|_1. \end{aligned} \quad (47)$$

The second inequality uses contractivity under the channels $\mathcal{A}_n \circ \mathcal{D}_n \circ \mathcal{E}_n$ and $\mathcal{A}_n$ in the first and second terms, respectively. The middle term in the final bound averages to $\bar{\delta}_n$, while the first and last terms sum to at most e_n for every g . Haar averaging gives

$$\int_{U(d)} \frac{1}{2} \|\mathcal{A}_n \circ \mathcal{D}_n \circ \mathcal{E}_n \circ \mathcal{B}_n(\tau_{g,n}) - \tau_{g,n}\|_1 dg \leq \bar{\delta}_n + e_n.$$

Proposition 4 and Lemma 3 therefore imply, for sufficiently large n ,

$$|M_n| \geq \log \dim \mathcal{H}_\Lambda + \log \left(1 - \frac{\bar{\delta}_n + e_n}{\gamma_x} \right) = \log \dim \mathcal{H}_\Lambda + o(1).$$

Finally, Lemma 1 applied to $\Lambda_i = nx_i + O(\sqrt{n} \log n)$ for $i \leq r$ and $\Lambda_i = 0$ for $i > r$ gives $\log \dim \mathcal{H}_\Lambda = L_{d,r}(n, x) + o(1)$. $\square$

6 Universal quantum data compression and QMDL

The QMDL memory cost also determines the extra average code length required by an unknown eigenbasis in universal lossless compression. For a fixed spectrum $x_1 > \dots > x_r > 0$, let $R_n(x)$ be the smallest worst-case excess in ideal average length above $nS(\rho)$ among codes that work for every eigenbasis. We will show that

$$R_n(x) = L_{d,r}(n, x) - s_{\text{blk}}(x) + o(1), \quad (48)$$

where $L_{d,r}(n, x)$ is the optimal QMDL cost from Theorem 1, and $s_{\text{blk}}(x)$ is the finite, nonnegative entropy in (51). Thus the optimal ideal average length is $nS(\rho) + R_n(x)$.

To make the coding criterion precise, a normalized code state σ_n , with support containing that of $\rho^{\otimes n}$, defines the ideal mean length $-\text{Tr} \rho^{\otimes n} \log \sigma_n$. Its excess over $nS(\rho)$ is $D(\rho^{\otimes n} \|\sigma_n)$. The quantum Kraft inequality gives a prefix code with lengths $\lceil -\log s_j \rceil$ in an eigenbasis of σ_n , where s_j are its nonzero eigenvalues [35, 36, 37].

For known x , the code may depend on the spectrum. The Haar-averaged source state is

$$\bar{\rho}_n := \int_{U(d)} \rho_g^{\otimes n} dg.$$

More generally, a distribution P_n on partitions $\lambda \vdash n$ with at most d rows defines the invariant code state

$$\sigma_{P_n, n} := \bigoplus_{\lambda} P_n(\lambda) \frac{\mathbb{I}_{\mathcal{H}_\lambda}}{\dim \mathcal{H}_\lambda} \otimes \tau_\lambda.$$

For brevity, write $q(\lambda) := q_{\lambda,n}$ for the source's Schur-label distribution. Schur–Weyl duality gives $\bar{\rho}_n = \sigma_{q,n}$. The next proposition identifies the optimal code state and separates the cost of an unknown eigenbasis from the cost of choosing a different label distribution.

Proposition 5 (Optimal overhead for an unknown eigenbasis). *For any density matrix ρ and any n , the Haar average $\bar{\rho}_n$ is an exact minimax code state:*

$$R_n(x) := \inf_{\sigma_n} \sup_{g \in \mathcal{U}(d)} D(\rho_g^{\otimes n} \| \sigma_n) = D(\rho^{\otimes n} \| \bar{\rho}_n) = \mathbb{E}_\lambda[\log \dim \mathcal{H}_\lambda - S(\rho_\lambda)]. \quad (49)$$

For every invariant code state $\sigma_{P_n,n}$,

$$D(\rho^{\otimes n} \| \sigma_{P_n,n}) = R_n(x) + D(q \| P_n), \quad (50)$$

with value $+\infty$ if the support condition fails.

Proof. For any state σ_n , Haar averaging gives

$$\int D(\rho_g^{\otimes n} \| \sigma_n) dg = D(\bar{\rho}_n \| \sigma_n) + S(\bar{\rho}_n) - nS(\rho) = D(\bar{\rho}_n \| \sigma_n) + D(\rho^{\otimes n} \| \bar{\rho}_n).$$

Thus the worst-case redundancy is at least $D(\rho^{\otimes n} \| \bar{\rho}_n)$. The invariant state $\bar{\rho}_n$ attains this value for every g . Finally, the Schur–Weyl block decomposition gives

$$D(\rho^{\otimes n} \| \sigma_{P_n,n}) = D(q \| P_n) + \sum_{\lambda} q(\lambda) [\log \dim \mathcal{H}_\lambda - S(\rho_\lambda)] = D(q \| P_n) + R_n(x).$$

□

Haar averaging replaces each ρ_λ by the maximally mixed state on $\mathcal{H}_\lambda$. Its entropy therefore increases by $\log \dim \mathcal{H}_\lambda - S(\rho_\lambda)$, the quantity averaged in (49). The average block entropy is already included in the source entropy $nS(\rho)$: redundancy counts only the additional uncertainty caused by the unknown eigenbasis. QMDL instead measures the memory dimension needed to recover the orbit, whose optimal logarithm is $L_{d,r}(n, x) + o(1)$. The offset $s_{\text{blk}}(x)$ is therefore the intrinsic mixedness of a typical representation state. The following lemma evaluates its limit and controls atypical blocks.

Lemma 4 (Block entropy). *Let ρ have strictly decreasing nonzero spectrum $x_1 > \dots > x_r > 0$. The entropies $S(\rho_\lambda)$ are bounded uniformly over all partitions λ with at most r rows. For $r \geq 2$, as $g_\lambda := \min_{i < r} (\lambda_i - \lambda_{i+1}) \rightarrow \infty$,*

$$S(\rho_\lambda) \longrightarrow s_{\text{blk}}(x),$$

where

$$s_{\text{blk}}(x) := \sum_{1 \leq i < j \leq r} \left[-\log \left(1 - \frac{x_j}{x_i} \right) - \frac{x_j}{x_i - x_j} \log \frac{x_j}{x_i} \right]. \quad (51)$$

For $r = 1$, every block entropy and the empty sum are zero.

Proof. See Appendix A. □

Each summand in (51) is the entropy of a geometric distribution with ratio x_j/x_i .

On the typical Schur diagrams, the supported row gaps grow linearly with n when $r \geq 2$. Lemmas 4 and 1 therefore give uniform limits for the block entropy and log dimension. The atypical contribution is $o(1)$ by (42), the uniform entropy bound, and $\log \dim \mathcal{H}_\lambda \leq \binom{d}{2} \log(n+1)$. Consequently,

$$\mathbb{E}_\lambda[S(\rho_\lambda)] = s_{\text{blk}}(x) + o(1), \quad \mathbb{E}_\lambda[\log \dim \mathcal{H}_\lambda] = L_{d,r}(n, x) + o(1).$$

Substitution in (49) proves the claimed relation (48) between QMDL and coding overhead.

When the spectrum is also unknown, the exact split (50) adds the classical penalty $D(q \| P_n)$. Appendix B evaluates this additional cost for Hayashi's universal codes.

7 Trace-distance bounds for generalized cloning

We now prove Theorem 2. We retain the highest-weight Kraus branch of each channel and use the Casimir gap to bound its deficit in each weight block. The mean depth is bounded uniformly over all representations, so averaging these deficits gives a finite error bound without truncating the states. The remaining factors are a scalar eigenvalue ratio and the ratio of Weyl dimensions. Figure 2 shows how these estimates combine for the forward channel, using the weight projectors P_δ, Q_δ defined in Section 7.1. By covariance it suffices to take $\rho = \text{diag}(x_1, \dots, x_r, 0, \dots, 0)$ throughout this section.

7.1 Technical preliminaries

We retain the assumptions and notation $\mu, \nu, \omega, D, b_\mu$ of Theorem 2; in particular, $\nu = \mu + \omega$ and $D = \|\omega\|_1 = \sum_i |\omega_i|$. Recall that $d_\lambda = \dim \mathcal{H}_\lambda$. For operators, $\|\cdot\|$ denotes the operator norm and $\|\cdot\|_1$ the trace norm; $X \geq Y$ means that $X - Y$ is positive semidefinite. For Hermitian X , its positive part X_+ is obtained by replacing each negative eigenvalue by zero.

We use the standard basis e_i and the Euclidean inner product on weights [38]. The positive roots of the first s coordinates are $\Phi_s^+ := \{e_i - e_j : 1 \leq i < j \leq s\}$, and the simple roots are $\alpha_i = e_i - e_{i+1}$. The Weyl group S_d acts by permuting coordinates. Write

$$Q_+ := \sum_{i=1}^{d-1} \mathbb{Z}_{\geq 0} \alpha_i, \quad Q_+^{(r)} := \sum_{i=1}^{r-1} \mathbb{Z}_{\geq 0} \alpha_i.$$

For $\delta = \sum_i c_i \alpha_i$, its depth is $|\delta| := \sum_i c_i$. Thus $|e_i - e_j| = j - i$ for $i < j$. This depth is distinct from the coordinate ℓ^1 norm used in D . Here δ denotes a weight offset; the compression error is denoted by δ_n .

Every weight of $\mathcal{H}_\lambda$ is of the form $\lambda - \delta$ with $\delta \in Q_+$. Let $\Pi_{\lambda-\delta}$ project onto this weight space, with multiplicity $m_\lambda(\delta) := \text{rank } \Pi_{\lambda-\delta}$. We take the multiplicity and projector to be zero when the weight is absent. The highest-weight space is one-dimensional; write $|\lambda\rangle$ for a unit vector spanning it.

The Cartan generator H_i and root generator $E_{e_i - e_j}$ are the represented matrix units E_{ii} and E_{ij} , respectively. On a vector of weight w , H_i acts by w_i , while E_α maps it to weight $w + \alpha$ or to zero. In particular, $E_\alpha |\lambda\rangle = 0$ for $\alpha > 0$. We use the invariant inner product for which $H_i^\dagger = H_i$ and $E_\alpha^\dagger = E_{-\alpha}$.

Let $P_s(\delta)$ count the expressions of δ as a nonnegative integer combination of roots in Φ_s^+ , with value zero when no such expression exists. This is Kostant's partition function. Ordered products of lowering operators applied to $|\lambda\rangle$ span the representation, giving the Verma-module bound $m_\lambda(\delta) \leq P_d(\delta)$. For $\delta \in Q_+^{(r)}$, positive roots outside the first r coordinates cannot occur in a sum with all other simple-root coefficients zero. Hence

$$m_\lambda(\delta) \leq P_d(\delta) = P_r(\delta), \quad \delta \in Q_+^{(r)}. \quad (52)$$

Lemma 5 (Multiplicity monotonicity). *Let μ, ω be dominant integral highest weights of $\text{GL}(d, \mathbb{C})$. Then $m_\mu(\delta) \leq m_{\mu+\omega}(\delta)$ for every $\delta \in Q_+$.*

Proof. Weight multiplicities count integral Gelfand–Tsetlin patterns $(m_{i,j})_{1 \leq j \leq i \leq d}$, with top row $m_{d,j} = \mu_j$ and interlacing inequalities $m_{i+1,j} \geq m_{i,j} \geq m_{i+1,j+1}$. Their weight is the vector of successive row-sum differences. The map $m_{i,j} \mapsto m_{i,j} + \omega_j$ is injective, changes the top row from μ to $\mu + \omega$, and shifts the weight by ω . It preserves interlacing: one inequality is unchanged, while the other gains $\omega_j - \omega_{j+1} \geq 0$. Thus it maps patterns counted by $m_\mu(\delta)$ into those counted by $m_{\mu+\omega}(\delta)$. $\square$

For $\rho = \text{diag}(x_1, \dots, x_r, 0, \dots, 0)$ and a partition λ with at most r rows,

$$\rho_\lambda = \sum_{\delta \in Q_+^{(r)}} p_\lambda(\delta) \Pi_{\lambda-\delta}, \quad p_\lambda(\delta) := \frac{x^{\lambda-\delta}}{s_\lambda(x)}, \quad x^w := \prod_{i=1}^r x_i^{w_i}.$$

Only these offsets contribute: both λ and each supported weight have zero coordinates after r , so their difference uses only $\alpha_1, \dots, \alpha_{r-1}$. The scalar $p_\lambda(\delta)$ is defined even when the weight is absent; on each nonzero weight space it is the eigenvalue of ρ_λ . The block probabilities are $p_\lambda(\delta)m_\lambda(\delta)$ and sum to one. The normalizer $s_\lambda(x) = \text{Tr} \pi_\lambda(\rho)$ is the Schur polynomial [39]. All sums over offsets below are over $Q_+^{(r)}$, unless stated otherwise.

For $r \geq 2$, set

$$g_\lambda := \min_{1 \leq i < r} (\lambda_i - \lambda_{i+1}).$$

This measures the gaps within the first r rows. In contrast, b_μ from Theorem 2 also includes the gap to the zero row when $r < d$:

$$b_\mu = \begin{cases} \min\{g_\mu, \mu_r\}, & 2 \leq r < d, \\ g_\mu, & r = d. \end{cases}$$

Thus $b_\mu \leq g_\mu$, while dominance of ω gives $g_\nu \geq g_\mu$. For $r = 1$, $b_\mu = \mu_1$, the only supported offset is $\delta = 0$, and no g_λ is needed.

On $\mathcal{H}_\mu \otimes \mathcal{H}_\omega$, each Lie-algebra generator acts as $X^{(\mu)} \otimes \mathbb{I}_\omega + \mathbb{I}_\mu \otimes X^{(\omega)}$, so tensor-product weights add. The vector $|\mu\rangle \otimes |\omega\rangle$ spans the one-dimensional weight space of weight $\nu = \mu + \omega$ and generates the multiplicity-one Cartan component. Choose the phase of the isometry V in Proposition 1 so that $V|\nu\rangle = |\mu\rangle \otimes |\omega\rangle$. Every other irreducible constituent has highest weight $\chi = \nu - \eta$ with $0 \neq \eta \in Q_+$.

For each supported offset δ , define two projectors on $\mathcal{H}_\mu \otimes \mathcal{H}_\omega$:

$$P_\delta := V \Pi_{\nu-\delta} V^\dagger, \quad Q_\delta := \Pi_{\mu-\delta} \otimes |\omega\rangle\langle\omega|. \quad (53)$$

The projector P_δ selects the embedded target weight space, while Q_δ selects the input weight space with the auxiliary system fixed to its highest weight. Both are supported on the full tensor-product weight space $\mathcal{W}_{\nu-\delta}$ of weight $\nu - \delta$. Since V intertwines the Cartan action, P_δ is the restriction of $VV^\dagger$ to this space. For Q_δ , fixing the auxiliary weight to ω forces the input weight to be $\mu - \delta$.

The quadratic Casimir uses the generators of the ambient $\text{GL}(d)$, even when the supported offsets lie in $Q_+^{(r)}$:

$$C_2 = \sum_i H_i^2 + \sum_{\alpha > 0} (E_{-\alpha} E_\alpha + E_\alpha E_{-\alpha}).$$

It acts on $\mathcal{H}_\lambda$ by $c_\lambda = (\lambda, \lambda + 2\varrho)$, where $\varrho = \frac{1}{2} \sum_{\alpha \in \Phi_d^+} \alpha$ and $(\varrho, \alpha_i) = 1$. On the tensor product, the total Casimir therefore acts by c_χ on each constituent $\mathcal{H}_\chi$.

7.2 Proof

Stage 1: the highest-weight Kraus branch. Using the Cartan intertwiner $V : \mathcal{H}_\nu \rightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega$ from Proposition 1, embed the forward channel as

$$V \mathcal{C}_{\mu \rightarrow \nu}(X) V^\dagger = \frac{d_\mu}{d_\nu} P_\nu(X \otimes \mathbb{I}_\omega) P_\nu, \quad (54)$$

$$\begin{aligned}
\frac{1}{2}\|\mathcal{C}_{\mu\rightarrow\nu}(\rho_\mu) - \rho_\nu\|_1 &\leq \underbrace{\text{Tr}(V\rho_\nu V^\dagger - \mathcal{K}_\omega(\rho_\mu))_+}_{\text{Retaining the highest-weight Kraus operator (57)}} \\
&= \sum_\delta \text{Tr}\left[p_\nu(\delta)P_\delta - \underbrace{p_\mu(\delta)}_{\substack{\geq p_\nu(\delta) \\ \text{Lemma 9}}} \mathcal{K}_\omega(\Pi_{\mu-\delta})\right]_+ \\
&\leq \underbrace{\left(1 - \frac{d_\mu}{d_\nu}\right) + \frac{d_\mu}{d_\nu} \sum_\delta p_\nu(\delta) \text{Tr}[P_\delta(\mathbb{I} - Q_\delta)P_\delta]}_{\text{Insert the Kraus formula (57)}} \\
&\leq \underbrace{\left(1 - \frac{d_\mu}{d_\nu}\right)}_{\substack{\text{Dimension ratio} \\ \text{Lemma 10}}} + \underbrace{\frac{2D}{g_\mu + 2}}_{\substack{\text{Projector estimate} \\ \text{Lemmas 6-7}}} \underbrace{\sum_\delta |\delta| p_\nu(\delta) m_\nu(\delta)}_{\substack{\text{Mean depth} \\ \text{Lemma 8}}} \\
&\leq \binom{d}{2} \frac{D}{b_\mu + 1} + \frac{2D}{g_\mu + 2} K(x) \leq C_{d,x} \frac{D}{b_\mu + 1}.
\end{aligned}$$

Figure 2: Contributions to the forward trace-distance bound. The reverse-channel proof is analogous.

where $P_\nu = VV^\dagger \in \mathcal{B}(\mathcal{H}_\mu \otimes \mathcal{H}_\omega)$ is the projector onto the embedded copy of $\mathcal{H}_\nu$. The embedded target state is

$$\tilde{\rho}_\nu := V\rho_\nu V^\dagger = \sum_\delta p_\nu(\delta)P_\delta. \quad (55)$$

Since the trace norm is invariant under a common isometric embedding,

$$\|\mathcal{C}_{\mu\rightarrow\nu}(\rho_\mu) - \rho_\nu\|_1 = \|V\mathcal{C}_{\mu\rightarrow\nu}(\rho_\mu)V^\dagger - \tilde{\rho}_\nu\|_1. \quad (56)$$

The highest-weight Kraus operator is

$$\mathcal{K}_\omega(X) := \frac{d_\mu}{d_\nu} P_\nu(X \otimes |\omega\rangle\langle\omega|)P_\nu. \quad (57)$$

Since $\mathbb{I}_\omega \geq |\omega\rangle\langle\omega|$, positivity gives

$$V\mathcal{C}_{\mu\rightarrow\nu}(\rho_\mu)V^\dagger \geq \mathcal{K}_\omega(\rho_\mu) = \sum_\delta p_\mu(\delta)\mathcal{K}_\omega(\Pi_{\mu-\delta}). \quad (58)$$

The omitted branches form a positive operator of trace $1 - \text{Tr}\mathcal{K}_\omega(\rho_\mu)$. The triangle inequality and isometric invariance therefore give

$$\frac{1}{2}\|\mathcal{C}_{\mu\rightarrow\nu}(\rho_\mu) - \rho_\nu\|_1 \leq \frac{1}{2}\|\mathcal{K}_\omega(\rho_\mu) - \tilde{\rho}_\nu\|_1 + \frac{1 - \text{Tr}\mathcal{K}_\omega(\rho_\mu)}{2} = \text{Tr}(\tilde{\rho}_\nu - \mathcal{K}_\omega(\rho_\mu))_+. \quad (59)$$

Each $\mathcal{K}_\omega(\Pi_{\mu-\delta})$ is supported on the total-weight sector $\nu - \delta$, so the branch can be compared with the target block by block.

Stage 2: estimating the projectors. We first identify the dimensions of the shallow weight spaces, then use the spectral gap of the total Casimir to compare their projectors.

Lemma 6 (Shallow weight multiplicities). *Let λ be a dominant integral highest weight of $\mathrm{GL}(d, \mathbb{C})$ and fix $2 \leq r \leq d$. Set $g_\lambda := \min_{1 \leq i < r} (\lambda_i - \lambda_{i+1})$. For $\delta \in Q_+^{(r)}$ with $|\delta| \leq g_\lambda$,*

$$m_\lambda(\delta) = P_r(\delta),$$

with P_r as defined in Section 7.1.

Proof. See Appendix A. □

Lemma 7 (Highest-weight subspace perturbation). *Let μ, ν be partitions with at most r rows and let $\omega = \nu - \mu$ be a dominant integral highest weight of $\mathrm{GL}(d, \mathbb{C})$. For $\delta \in Q_+^{(r)}$, let P_δ, Q_δ be the projectors in (53). Set $D := \|\omega\|_1$. For $r \geq 2$, set $g_\mu := \min_{1 \leq i < r} (\mu_i - \mu_{i+1})$ and assume $|\delta| \leq g_\mu$. Then*

$$\mathrm{rank} P_\delta = \mathrm{rank} Q_\delta = m_\mu(\delta)$$

and, for $r \geq 2$,

$$\|P_\delta - Q_\delta\| \leq \sqrt{\frac{2(\delta, \omega)}{g_\mu + 2}} \leq \sqrt{\frac{2|\delta|D}{g_\mu + 2}}. \quad (60)$$

When $r = 1$, only $\delta = 0$ occurs and the two projectors coincide.

Proof. If $|\delta|D = 0$, the two projectors coincide. Otherwise $r \geq 2$ and $|\delta| \leq g_\mu$ by hypothesis. Work on the total-weight space $\mathcal{W}_{\nu-\delta}$. Since ω is dominant, $g_\nu \geq g_\mu$. Lemma 6 therefore gives

$$\mathrm{rank} P_\delta = m_\nu(\delta) = P_r(\delta) = m_\mu(\delta) = \mathrm{rank} Q_\delta.$$

Restrict the total quadratic Casimir C_2 to this weight space. A constituent $\mathcal{H}_\chi$ of $\mathcal{H}_\mu \otimes \mathcal{H}_\omega$ that meets $\mathcal{W}_{\nu-\delta}$ has highest weight $\chi = \nu - \eta$ with $\eta, \delta - \eta \in Q_+$. Thus η is supported on the first $r - 1$ simple roots. For $\chi \neq \nu$, write $\eta = \sum_{i < r} b_i \alpha_i$ with $\sum_i b_i = |\eta| \geq 1$. Dominance of χ and the supported row gaps of ν give

$$\begin{aligned} c_\nu - c_\chi &= (\nu + \chi + 2\rho, \eta) \\ &= \sum_{i < r} b_i [(\nu_i - \nu_{i+1}) + (\chi_i - \chi_{i+1}) + 2] \\ &\geq (g_\mu + 2)|\eta| \geq g_\mu + 2. \end{aligned} \quad (61)$$

The Cartan component occurs once, so

$$c_\nu \mathbb{I} - C_2 \geq (g_\mu + 2)(\mathbb{I} - P_\delta).$$

It remains to evaluate the Casimir deficit on Q_δ . In the tensor-product Casimir, the cross terms are

$$2 \sum_i H_i^{(\mu)} \otimes H_i^{(\omega)} + 2 \sum_{\alpha > 0} (E_{-\alpha}^{(\mu)} \otimes E_\alpha^{(\omega)} + E_\alpha^{(\mu)} \otimes E_{-\alpha}^{(\omega)}).$$

The root-exchange terms vanish when compressed by Q_δ , because the auxiliary register lies in its highest-weight line. The Cartan term is $2(\mu - \delta, \omega)Q_\delta$. Using $c_\nu = c_\mu + c_\omega + 2(\mu, \omega)$, we obtain

$$Q_\delta(c_\nu \mathbb{I} - C_2)Q_\delta = 2(\delta, \omega)Q_\delta. \quad (62)$$

Consequently,

$$\|(\mathbb{I} - P_\delta)Q_\delta\|^2 = \|Q_\delta(\mathbb{I} - P_\delta)Q_\delta\| \leq \frac{2(\delta, \omega)}{g_\mu + 2}.$$

The projectors have equal rank, so $\|P_\delta - Q_\delta\| = \|(\mathbb{I} - P_\delta)Q_\delta\|$.³ Finally, $0 \leq (\delta, \omega) \leq |\delta|D$ gives (60). $\square$

Stage 3: averaging the block bounds. The uniformly bounded mean depth controls both the averaged projector deficits and the tail mass needed to estimate the eigenvalue ratio.

Lemma 8 (Uniform mean depth). *Let ρ be a density matrix of rank $r \leq d$ with strictly decreasing nonzero spectrum $x_1 > \dots > x_r > 0$. Let μ be a partition with at most r rows. With q_x as in Lemma 3, including $q_x = 0$ for $r = 1$, set*

$$N := |\Phi_r^+| = \binom{r}{2}, \quad K(x) := \frac{Nq_x}{(1 - q_x)^{N+1}}.$$

Then, uniformly over μ ,

$$\sum_{\delta} |\delta| p_{\mu}(\delta) m_{\mu}(\delta) \leq K(x). \quad (63)$$

For $r = 1$, both the mean depth and $K(x)$ are zero.

Proof. For $r = 1$, only the highest-weight line is supported, so the mean depth vanishes. Assume $r \geq 2$. Since $s_{\mu}(x) \geq x^{\mu}$ and $x^{-\delta} = \prod_{i < r} (x_{i+1}/x_i)^{c_i}$ for $\delta = \sum_{i < r} c_i \alpha_i$,

$$p_{\mu}(\delta) \leq q_x^{|\delta|}.$$

The multiplicity bound (52) gives

$$\sum_{|\delta|=t} m_{\mu}(\delta) \leq [z^t] \prod_{\alpha \in \Phi_r^+} (1 - z^{|\alpha|})^{-1} \leq [z^t] (1 - z)^{-N} = \binom{t + N - 1}{N - 1}.$$

Here $[z^t]$ denotes the coefficient of z^t . Differentiating $(1 - q_x)^{-N}$ now gives

$$\sum_{\delta} |\delta| p_{\mu}(\delta) m_{\mu}(\delta) \leq \sum_{t \geq 0} t \binom{t + N - 1}{N - 1} q_x^t = \frac{Nq_x}{(1 - q_x)^{N+1}} = K(x).$$

$\square$

The eigenvalue ratio $p_{\nu}(\delta)/p_{\mu}(\delta)$ is independent of δ . Its deviation from one is controlled by the weight sectors where the multiplicities can differ.

Lemma 9 (Eigenvalue ratio). *Let μ, ν be partitions with at most r rows and suppose $\omega := \nu - \mu$ is dominant. For a strictly decreasing positive spectrum $x_1 > \dots > x_r > 0$, set*

$$Z(x) := \frac{x^{\omega} s_{\mu}(x)}{s_{\nu}(x)} = \frac{p_{\nu}(\delta)}{p_{\mu}(\delta)}.$$

If $r = 1$, then $Z(x) = 1$. If $r \geq 2$, then

$$0 \leq 1 - Z(x) \leq \frac{K(x)}{g_{\mu} + 1}.$$

³For projectors P, Q , the identity $(P - Q)^2 = P(\mathbb{I} - Q)P + (\mathbb{I} - P)Q(\mathbb{I} - P)$ is block diagonal with respect to $\text{ran } P \oplus \ker P$, and therefore $\|P - Q\| = \max\{\|(\mathbb{I} - Q)P\|, \|(\mathbb{I} - P)Q\|\}$. When the ranks agree, PQP on $\text{ran } P$ and QPQ on $\text{ran } Q$ have the same eigenvalues: they are the squared singular values of $PQ : \text{ran } Q \rightarrow \text{ran } P$. Thus $P(\mathbb{I} - Q)P$ and $Q(\mathbb{I} - P)Q$ have equal norms, so the two terms in the maximum coincide.

Proof. Normalization and Lemma 5 give

$$1 - Z(x) = \sum_{\delta} p_{\nu}(\delta) [m_{\nu}(\delta) - m_{\mu}(\delta)] \geq 0.$$

For $r \geq 2$, the multiplicities agree at depths $|\delta| \leq g_{\mu}$ by Lemma 6, since $g_{\nu} \geq g_{\mu}$. Since $0 \leq m_{\nu}(\delta) - m_{\mu}(\delta) \leq m_{\nu}(\delta)$, the remaining sum satisfies

$$1 - Z(x) = \sum_{|\delta| > g_{\mu}} p_{\nu}(\delta) [m_{\nu}(\delta) - m_{\mu}(\delta)] \leq \sum_{|\delta| > g_{\mu}} p_{\nu}(\delta) m_{\nu}(\delta).$$

Both the depth and g_{μ} are integers, so $|\delta| > g_{\mu}$ implies $|\delta| \geq g_{\mu} + 1$. Therefore

$$\begin{aligned} \sum_{|\delta| > g_{\mu}} p_{\nu}(\delta) m_{\nu}(\delta) &\leq \frac{1}{g_{\mu} + 1} \sum_{|\delta| > g_{\mu}} |\delta| p_{\nu}(\delta) m_{\nu}(\delta) \\ &\leq \frac{1}{g_{\mu} + 1} \sum_{\delta} |\delta| p_{\nu}(\delta) m_{\nu}(\delta) \leq \frac{K(x)}{g_{\mu} + 1}, \end{aligned}$$

where the last step is Lemma 8, Eq. (63), applied to ν . For $r = 1$, only $\delta = 0$ occurs and both multiplicities are one. $\square$

The other scalar factor is the ratio of representation dimensions.

Lemma 10 (Weyl dimension ratio). *Under the partition and dominance assumptions of Theorem 2,*

$$0 \leq 1 - \frac{d_{\mu}}{d_{\nu}} \leq \sum_{i < j} \frac{\omega_i - \omega_j}{\mu_i - \mu_j + j - i} \leq \binom{d}{2} \frac{D}{b_{\mu} + 1}.$$

Proof. The Weyl dimension formula gives

$$\frac{d_{\mu}}{d_{\nu}} = \prod_{i < j} \left(1 - \frac{\omega_i - \omega_j}{\nu_i - \nu_j + j - i} \right).$$

Each fraction lies in $[0, 1)$. Using $1 - \prod_k (1 - a_k) \leq \sum_k a_k$ for $a_k \in [0, 1]$ yields

$$1 - \frac{d_{\mu}}{d_{\nu}} \leq \sum_{i < j} \frac{\omega_i - \omega_j}{\nu_i - \nu_j + j - i} \leq \sum_{i < j} \frac{\omega_i - \omega_j}{\mu_i - \mu_j + j - i}.$$

For $i > r$, the numerator is zero. For $i \leq r$, it is at most D , while the denominator is at least $b_{\mu} + 1$. $\square$

We are now ready to complete the proof of Theorem 2.

Proof of Theorem 2. If $D = 0$, then $\omega = 0$ and both channels are the identity by Propositions 1 and 2. Hence assume $D \geq 1$, as D is an integer. We first bound the projector deficits at every depth, then average them and compare the two channel outputs with their target states. All sums below run over $\delta \in Q_+^{(r)}$; absent weight spaces have zero projector and multiplicity, as in Section 7.1.

Projector deficits and their averages. For $r \geq 2$, set

$$e_{\delta} := \min \left\{ 1, \frac{2|\delta|D}{g_{\mu} + 2} \right\}.$$

At depths $|\delta| \leq g_\mu$, Lemma 7, together with the general bound $\|P_\delta - Q_\delta\| \leq 1$ for projectors, gives $\|P_\delta - Q_\delta\|^2 \leq e_\delta$. At greater depths, integrality gives $|\delta| \geq g_\mu + 1$, so

$$\frac{2|\delta|D}{g_\mu + 2} \geq \frac{2(g_\mu + 1)}{g_\mu + 2} \geq 1.$$

Thus $e_\delta = 1$ and the same projector bound holds without any assumption on the ranks of the two weight spaces. For $r = 1$, only $\delta = 0$ contributes; Lemma 7 gives $P_0 = Q_0$, and we set $e_0 = 0$.

The norm estimate controls the positive operator measuring the part of P_δ outside Q_δ :

$$\|P_\delta(\mathbb{I} - Q_\delta)P_\delta\| = \|(\mathbb{I} - Q_\delta)P_\delta\|^2 \leq \|P_\delta - Q_\delta\|^2 \leq e_\delta.$$

Since this operator is supported on P_δ , it is at most $e_\delta P_\delta$. Interchanging the projectors gives the analogous bound on Q_δ , and hence

$$P_\delta Q_\delta P_\delta \geq (1 - e_\delta)P_\delta, \quad Q_\delta P_\delta Q_\delta \geq (1 - e_\delta)Q_\delta. \quad (64)$$

The cap $e_\delta \leq 1$ ensures that both lower bounds have nonnegative coefficients. By (53), $\text{Tr}P_\delta = m_\nu(\delta)$ and $\text{Tr}Q_\delta = m_\mu(\delta)$. For $r \geq 2$, the corresponding averaged trace deficits are controlled by

$$M_\lambda := \sum_\delta e_\delta p_\lambda(\delta) m_\lambda(\delta) \leq \frac{2D}{g_\mu + 2} \sum_\delta |\delta| p_\lambda(\delta) m_\lambda(\delta) \leq \frac{2DK(x)}{g_\mu + 2}, \quad \lambda \in \{\mu, \nu\}, \quad (65)$$

where the last inequality is Lemma 8, Eq. (63). Its constant is uniform over both partitions. For $r = 1$, define $M_\mu = M_\nu = 0$ instead.

Equation (59) reduces the forward error to the trace-distance term for the retained branch together with its missing trace. We will control both terms at once by constructing a positive remainder R_f such that

$$\mathcal{K}_\omega(\rho_\mu) \geq \tilde{\rho}_\nu - R_f.$$

The following elementary inequality shows that $\text{Tr}R_f$ then bounds the entire right-hand side of (59). Let $A \geq 0$ have $\text{Tr}A \leq 1$, let B be a state, and suppose $A \geq B - R$ for some $R \geq 0$. Since $\text{Tr}(B - A) = 1 - \text{Tr}A$, the positive-part identity and its variational formula give

$$\frac{\|A - B\|_1 + 1 - \text{Tr}A}{2} = \text{Tr}(B - A)_+ = \max_{0 \leq L \leq \mathbb{I}} \text{Tr}[L(B - A)] \leq \text{Tr}R. \quad (66)$$

Indeed, $B - A \leq R$ implies $\text{Tr}[L(B - A)] \leq \text{Tr}(LR) \leq \text{Tr}R$ for every such L . For normalized A , the left-hand side is the usual trace distance; for a retained Kraus branch, it also includes the missing trace.

Forward channel. Lemma 9 gives $p_\nu(\delta) = Z(x)p_\mu(\delta)$ with $0 < Z(x) \leq 1$, so $p_\mu(\delta) \geq p_\nu(\delta)$. Lemma 10 gives $0 < d_\mu/d_\nu \leq 1$. Since Q_δ lies in the total-weight sector $\nu - \delta$, Eqs. (57) and (53) give

$$\mathcal{K}_\omega(\Pi_{\mu-\delta}) = \frac{d_\mu}{d_\nu} P_\delta Q_\delta P_\delta.$$

To construct the required remainder, use positivity of each term and the eigenvalue comparison to obtain

$$\mathcal{K}_\omega(\rho_\mu) = \frac{d_\mu}{d_\nu} \sum_\delta p_\mu(\delta) P_\delta Q_\delta P_\delta \geq \frac{d_\mu}{d_\nu} \sum_\delta p_\nu(\delta) P_\delta Q_\delta P_\delta = \tilde{\rho}_\nu - R_f,$$

where we used the target decomposition (55) and set

$$R_f := \left(1 - \frac{d_\mu}{d_\nu}\right) \tilde{\rho}_\nu + \frac{d_\mu}{d_\nu} \sum_\delta p_\nu(\delta) P_\delta (\mathbb{I} - Q_\delta) P_\delta \geq 0.$$

These two terms account for the dimension ratio and the mismatch between weight spaces, respectively. Taking traces and using (64) gives

$$\begin{aligned} \text{Tr} R_f &= 1 - \frac{d_\mu}{d_\nu} + \frac{d_\mu}{d_\nu} \sum_\delta p_\nu(\delta) \text{Tr}[P_\delta (\mathbb{I} - Q_\delta) P_\delta] \\ &\leq 1 - \frac{d_\mu}{d_\nu} + \frac{d_\mu}{d_\nu} \sum_\delta e_\delta p_\nu(\delta) m_\nu(\delta) \leq 1 - \frac{d_\mu}{d_\nu} + M_\nu. \end{aligned}$$

By (58), the retained branch has trace at most one. We may therefore apply (66) with $A = \mathcal{K}_\omega(\rho_\mu)$, $B = \tilde{\rho}_\nu$, and $R = R_f$. The two terms on its left are exactly the upper bound for the full channel in (59), including the trace of the omitted branches. Consequently,

$$\frac{1}{2} \|\mathcal{C}_{\mu \rightarrow \nu}(\rho_\mu) - \rho_\nu\|_1 \leq \text{Tr} R_f \leq 1 - \frac{d_\mu}{d_\nu} + M_\nu. \quad (67)$$

Reverse channel. Taking the adjoint in Proposition 1 and using Proposition 2 cancels the dimension factors:

$$\mathcal{C}_{\nu \rightarrow \mu}(X) = \frac{d_\nu}{d_\mu} \mathcal{C}_{\mu \rightarrow \nu}^\dagger(X) = \text{Tr}_\omega(V X V^\dagger).$$

Define the isometry $Jv := v \otimes |\omega\rangle$. The partial trace is a sum of positive matrix elements over an orthonormal auxiliary basis containing $|\omega\rangle$. Retaining that one term gives $\mathcal{C}_{\nu \rightarrow \mu}(\rho_\nu) \geq J^\dagger \tilde{\rho}_\nu J$. Moreover, in the total-weight sector $\nu - \delta$, fixing the auxiliary weight to ω forces the input weight to be $\mu - \delta$. Thus (53) implies

$$P_\delta J = P_\delta Q_\delta J, \quad J^\dagger Q_\delta J = \Pi_{\mu-\delta}.$$

Using these identities, the decomposition (55), and the second bound in (64), we obtain

$$\mathcal{C}_{\nu \rightarrow \mu}(\rho_\nu) \geq \sum_\delta p_\nu(\delta) J^\dagger Q_\delta P_\delta Q_\delta J \geq \sum_\delta p_\nu(\delta) (1 - e_\delta) \Pi_{\mu-\delta} = \sum_\delta Z(x) (1 - e_\delta) p_\mu(\delta) \Pi_{\mu-\delta},$$

where the last equality is Lemma 9. Since $0 < Z(x) \leq 1$ and $e_\delta \geq 0$, $Z(x)(1 - e_\delta) \geq Z(x) - e_\delta$. The output is therefore at least $\rho_\mu - R_r$, where

$$R_r := \sum_\delta (1 - Z(x) + e_\delta) p_\mu(\delta) \Pi_{\mu-\delta} \geq 0.$$

The reverse output is a state, so applying (66) and using $\sum_\delta p_\mu(\delta) m_\mu(\delta) = 1$ yields

$$\frac{1}{2} \|\mathcal{C}_{\nu \rightarrow \mu}(\rho_\nu) - \rho_\mu\|_1 \leq \text{Tr} R_r = 1 - Z(x) + M_\mu. \quad (68)$$

Collecting the estimates. For $r \geq 2$, substitute (65) into (67)–(68). Lemma 10 bounds the forward dimension loss, while Lemma 9 bounds the reverse eigenvalue loss. Using $b_\mu \leq g_\mu$ and $D \geq 1$ gives explicitly

$$\begin{aligned} \frac{1}{2} \|\mathcal{C}_{\mu \rightarrow \nu}(\rho_\mu) - \rho_\nu\|_1 &\leq \binom{d}{2} \frac{D}{b_\mu + 1} + \frac{2DK(x)}{g_\mu + 2} \leq \left[\binom{d}{2} + 2K(x) \right] \frac{D}{b_\mu + 1}, \\ \frac{1}{2} \|\mathcal{C}_{\nu \rightarrow \mu}(\rho_\nu) - \rho_\mu\|_1 &\leq \frac{K(x)}{g_\mu + 1} + \frac{2DK(x)}{g_\mu + 2} \leq 3K(x) \frac{D}{b_\mu + 1}. \end{aligned}$$

For $r = 1$, Lemma 9 gives $Z(x) = 1$, and $M_\mu = M_\nu = 0$. Equations (67) and (68), together with Lemma 10, then give the forward bound $\binom{d}{2}D/(b_\mu + 1)$ and zero reverse error. Thus in all cases one may take $C_{d,x} = \binom{d}{2} + 3K(x)$, with $K(x)$ as in Lemma 8 (and $K(x) = 0$ for $r = 1$). $\square$

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Statements and Declarations

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Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Data availability

This theoretical study does not use empirical datasets. The mathematical derivations supporting the results are included in the article and its appendices.

Code availability

The Lean formalization of Theorems 1 and 2, verification scripts, and supporting documentation are publicly available in the [project repository](#).

Generative AI usage

The authors developed the connection between QMDL and free entropy and the use of the PRV channel to extend YCH compression to qudits. Generative AI assisted with proof exploration, mathematical revisions, and language editing. Work with ChatGPT-o1, o3, and o4-mini developed an initial strategy based on $GL(d)$ Clebsch–Gordan coefficients. Gemini 3 and 3.1 later suggested a projector-perturbation argument using the Davis–Kahan theorem. The final proof of Lemma 7 uses a direct Casimir argument suggested by ChatGPT 6, yielding the finite bound in Theorem 2. ChatGPT 6 and Fable 5.1 also assisted with manuscript review. Codex assisted with the development of the Lean formalization and associated verification scripts. The authors reviewed and revised all AI-assisted material and take responsibility for the mathematical arguments and the manuscript’s content.

A Proofs

Proof of Proposition 1. Let $|\Omega\rangle = \sum_K |\mu^*, K^*\rangle \otimes |\mu, K\rangle$ be the unnormalized maximally entangled vector. By definition, $J = (\text{id} \otimes \mathcal{C}_{\mu \rightarrow \nu})(|\Omega\rangle\langle\Omega|)$, so using

$$\mathcal{C}_{\mu \rightarrow \nu}(\sigma) = \frac{d_\mu}{d_\nu} V^\dagger (\sigma \otimes \mathbb{I}_\omega) V,$$

we get

$$J = \frac{d_\mu}{d_\nu} \sum_{K,L} \sum_W |\mu^*, K^*\rangle\langle\mu^*, L^*| \otimes \left(V^\dagger |\mu, K; \omega, W\rangle\langle\mu, L; \omega, W| V \right).$$

Define a linear map $\tilde{V} : \mathcal{H}_{\mu^*} \otimes \mathcal{H}_\nu \rightarrow \mathcal{H}_\omega$ by

$$\tilde{V}(|\mu^*, K^*\rangle \otimes |\nu, N\rangle) := (\langle\mu, K| \otimes \mathbb{I}_\omega) V |\nu, N\rangle.$$

Then its matrix elements satisfy

$$\langle\mu^*, K^*; \nu, N| \tilde{V}^\dagger \tilde{V} |\mu^*, L^*; \nu, N'\rangle = \sum_W \langle\nu, N| V^\dagger |\mu, K; \omega, W\rangle \langle\mu, L; \omega, W| V |\nu, N'\rangle.$$

Comparing with the expression above for J , we obtain

$$J = \frac{d_\mu}{d_\nu} \tilde{V}^\dagger \tilde{V}.$$

Contracting the μ factor of the intertwiner V gives the intertwiner $\tilde{V} : \mu^* \otimes \nu \rightarrow \omega$. Hence $\tilde{V}^\dagger \tilde{V}$ commutes with the representation $\mu^* \otimes \nu$, and its support lies in the ω -isotypic component.

The Cartan component has multiplicity one, so

$$\dim \text{Hom}_{U(d)}(\nu, \mu \otimes \omega) = \dim \text{Hom}_{U(d)}(\mu^* \otimes \nu, \omega) = 1.$$

Therefore ω occurs with multiplicity one in $\mu^* \otimes \nu$. If Π_ω denotes the projector onto that unique copy, then $\tilde{V}^\dagger \tilde{V}$ is proportional to Π_ω . Its trace is

$$\text{Tr}(\tilde{V}^\dagger \tilde{V}) = \sum_{N,K} \|(\langle\mu, K| \otimes \mathbb{I}_\omega) V |\nu, N\rangle\|^2 = \sum_N \|V |\nu, N\rangle\|^2 = \text{Tr}(V^\dagger V) = d_\nu,$$

since V is an isometry. Since $\text{Tr} \Pi_\omega = d_\omega$, this gives

$$\tilde{V}^\dagger \tilde{V} = \frac{d_\nu}{d_\omega} \Pi_\omega.$$

Substituting back into the Choi formula gives

$$J = \frac{d_\mu}{d_\nu} \tilde{V}^\dagger \tilde{V} = \frac{d_\mu}{d_\omega} \Pi_\omega.$$

Since $(\nu - \mu)^+ = \omega$, this is the required PRV Choi operator. $\square$

Proof of Proposition 2. Covariance makes $\mathcal{C}_{\mu \rightarrow \nu}(\sigma_\mu)$ invariant. Since U_ν is irreducible, Schur's lemma and trace preservation give

$$\mathcal{C}_{\mu \rightarrow \nu}(\sigma_\mu) = \frac{\mathbb{I}_\nu}{d_\nu}.$$

Therefore the Petz map with respect to σ_μ is

$$\mathcal{R}_{\sigma_\mu, \mathcal{C}_{\mu \rightarrow \nu}}(X) = \frac{1}{d_\mu} \mathcal{C}_{\mu \rightarrow \nu}^\dagger(d_\nu X) = \frac{d_\nu}{d_\mu} \mathcal{C}_{\mu \rightarrow \nu}^\dagger(X).$$

So it remains to identify the scaled adjoint with the reverse cloning channel.

Let $\text{vec} : \text{Hom}(\mathcal{H}_\mu, \mathcal{H}_\nu) \rightarrow \mathcal{H}_\mu^* \otimes \mathcal{H}_\nu$ be the canonical vectorization isometry. Choose Hilbert–Schmidt orthonormal operators $K_a \in \text{Hom}(\mathcal{H}_\mu, \mathcal{H}_\nu)$, $1 \leq a \leq d_\omega$, whose vectorizations form an orthonormal basis for the range of Π_ω . Then

$$J_\omega = \frac{d_\mu}{d_\omega} \sum_{a=1}^{d_\omega} |\text{vec}(K_a)\rangle\langle\text{vec}(K_a)|,$$

and hence

$$\mathcal{C}_{\mu \rightarrow \nu}(X) = \frac{d_\mu}{d_\omega} \sum_{a=1}^{d_\omega} K_a X K_a^\dagger.$$

Consequently,

$$\frac{d_\nu}{d_\mu} \mathcal{C}_{\mu \rightarrow \nu}^\dagger(Y) = \frac{d_\nu}{d_\omega} \sum_{a=1}^{d_\omega} K_a^\dagger Y K_a.$$

Now the span $\text{span}\{K_a\} \subset \text{Hom}(\mathcal{H}_\mu, \mathcal{H}_\nu)$ carries the irrep ω under the action

$$K \mapsto U_\nu(g) K U_\mu(g)^\dagger.$$

Taking Hilbert–Schmidt adjoints defines an antiunitary intertwiner

$$K \mapsto K^\dagger$$

from $\text{Hom}(\mathcal{H}_\mu, \mathcal{H}_\nu)$ to $\text{Hom}(\mathcal{H}_\nu, \mathcal{H}_\mu)$. Therefore $\text{span}\{K_a^\dagger\} \subset \text{Hom}(\mathcal{H}_\nu, \mathcal{H}_\mu)$ carries the dual irrep ω^* . Since

$$\omega^* = (-w_0\omega) = (\mu - \nu)^+,$$

this is exactly the PRV component for the reversed pair (ν, μ) .

The operators $K_a^\dagger$ remain Hilbert–Schmidt orthonormal, so the scaled adjoint has Choi operator $(d_\nu/d_{\omega^*})\Pi_{\omega^*}$, with $d_{\omega^*} = d_\omega$. This is the defining Choi operator of $\mathcal{C}_{\nu \rightarrow \mu}$, proving

$$\frac{d_\nu}{d_\mu} \mathcal{C}_{\mu \rightarrow \nu}^\dagger = \mathcal{C}_{\nu \rightarrow \mu}.$$

Combining this with the Petz formula above proves the claim. $\square$

Proof of Lemma 3. For $r = 1$, ρ_λ is pure and $\gamma = \gamma_x = 1$. Assume $r \geq 2$ and use the weight notation of Section 7.1. The nonzero spectrum agrees with that of the corresponding $\text{GL}(r)$ representation. The highest-weight line has eigenvalue $p_0 = x^\lambda/s_\lambda(x)$. Every other nonzero eigenvalue, at weight $\lambda - \delta$, divided by p_0 is

$$x^{-\delta} = \prod_{i=1}^{r-1} \left(\frac{x_{i+1}}{x_i} \right)^{c_i} \leq q_x, \quad \delta = \sum_{i=1}^{r-1} c_i \alpha_i, \quad c_i \in \mathbb{Z}_{\geq 0}, \quad \sum_i c_i > 0.$$

Since $q_x < 1$ and the highest-weight space is one-dimensional, the largest eigenvalue is simple and $p_1 \leq q_x p_0$.

It remains to bound p_0 below uniformly in λ . Expanding the Schur polynomial over weight spaces and applying (52) term by term gives

$$p_0^{-1} = \frac{s_\lambda(x)}{x^\lambda} = \sum_{\delta \in Q_+^{(r)}} m_\lambda(\delta) x^{-\delta} \leq \sum_{\delta \in Q_+^{(r)}} P_r(\delta) x^{-\delta}.$$

Here all terms are nonnegative. By definition, $P_r(\delta)$ counts the choices of nonnegative integers k_{ij} satisfying

$$\delta = \sum_{1 \leq i < j \leq r} k_{ij}(e_i - e_j).$$

For each such choice, the definition $x^w = \prod_{\ell=1}^r x_\ell^{w_\ell}$ gives $x^{-(e_i - e_j)} = x_j/x_i$, and hence

$$x^{-\delta} = \prod_{1 \leq i < j \leq r} \left(\frac{x_j}{x_i} \right)^{k_{ij}}.$$

Thus summing over all offsets and their decompositions is equivalent to summing independently over every k_{ij} , yielding

$$\sum_{\delta \in Q_+^{(r)}} P_r(\delta) x^{-\delta} = \prod_{1 \leq i < j \leq r} \sum_{k=0}^{\infty} \left(\frac{x_j}{x_i} \right)^k = \prod_{1 \leq i < j \leq r} \left(1 - \frac{x_j}{x_i} \right)^{-1}.$$

The geometric series converge because $x_j/x_i < 1$ for $i < j$. Taking reciprocals gives $p_0 \geq \prod_{i < j \leq r} (1 - x_j/x_i)$, and hence

$$\gamma = p_0 - p_1 \geq (1 - q_x) p_0 \geq (1 - q_x) \prod_{1 \leq i < j \leq r} \left(1 - \frac{x_j}{x_i} \right) = \gamma_x.$$

□

Proof of Lemma 4. Assume $r \geq 2$ and use the weight notation of Section 7.1. All sums below are over $\delta \in Q_+^{(r)}$. Set

$$Z_\lambda := \sum_{\delta} m_\lambda(\delta) x^{-\delta}, \quad B_\lambda := \sum_{\delta} m_\lambda(\delta) x^{-\delta} \log x^\delta.$$

The nonzero eigenvalues of ρ_λ are $x^{-\delta}/Z_\lambda$, with multiplicity $m_\lambda(\delta)$, so

$$S(\rho_\lambda) = \log Z_\lambda + B_\lambda/Z_\lambda.$$

The Verma-module bound (52) gives $m_\lambda(\delta) \leq P_r(\delta)$, where P_r is the positive-root partition function. Replacing m_λ by P_r defines the convergent sums Z_∞, B_∞ . Their product formulas are

$$Z_\infty = \prod_{i < j \leq r} \left(1 - \frac{x_j}{x_i} \right)^{-1}, \quad \frac{B_\infty}{Z_\infty} = \sum_{i < j \leq r} \frac{x_j}{x_i - x_j} \log \frac{x_i}{x_j}.$$

Both formulas follow by summing the independent nonnegative occupation numbers of the positive roots. Since $1 \leq Z_\lambda \leq Z_\infty$ and $0 \leq B_\lambda \leq B_\infty$, the entropies are uniformly bounded. By Lemma 6, $m_\lambda(\delta) = P_r(\delta)$ whenever $|\delta| \leq g_\lambda$. The omitted tails of the two convergent positive sums tend to zero as $g_\lambda \rightarrow \infty$. Hence $Z_\lambda \rightarrow Z_\infty$ and $B_\lambda \rightarrow B_\infty$, giving (51). The case $r = 1$ is immediate because ρ_λ is pure. □

Proof of Lemma 6. We show that at these depths all nonidentity terms in Kostant's multiplicity formula vanish, so the multiplicity is exactly the number of positive-root partitions of δ . The formula [40] reads

$$m_\lambda(\delta) = \sum_{w \in S_d} \text{sgn}(w) P_d(\delta - [\lambda + \varrho - w(\lambda + \varrho)]).$$

Recall that P_d is zero outside Q_+ : its argument must have nonnegative coefficients in the simple-root basis. Write $\delta = \sum_{i=1}^{d-1} c_i \alpha_i$. By hypothesis, $c_i \geq 0$, $c_i = 0$ for $i \geq r$, and $c_i \leq |\delta| \leq g_\lambda$ for every i .

Fix $w \neq \text{id}$, and let $k \leq d-1$ be the first coordinate moved by w . Then $w^{-1}(i) = i$ for $i < k$, while $w^{-1}(k) > k$. For a vector with zero coordinate sum, the coefficient of $\alpha_k = e_k - e_{k+1}$ in its simple-root expansion is the sum of its first k coordinates. Thus the α_k coefficient of $\lambda + \varrho - w(\lambda + \varrho)$ is

$$\begin{aligned} \sum_{i=1}^k [(\lambda + \varrho)_i - (w(\lambda + \varrho))_i] &= (\lambda + \varrho)_k - (\lambda + \varrho)_{w^{-1}(k)} \\ &\geq (\lambda + \varrho)_k - (\lambda + \varrho)_{k+1} = \lambda_k - \lambda_{k+1} + 1. \end{aligned}$$

Here dominance of λ and $\varrho_i - \varrho_{i+1} = 1$ make $\lambda + \varrho$ strictly decreasing, which justifies the inequality. If $k < r$, the last expression is at least $g_\lambda + 1 > c_k$. If $k \geq r$, it is at least one, whereas $c_k = 0$. In either case, subtracting this vector from δ produces a negative α_k coefficient. The argument of P_d is therefore outside Q_+ , and the term indexed by w vanishes. Only $w = \text{id}$ remains, giving $m_\lambda(\delta) = P_d(\delta)$.

Finally, each positive root has the expansion $e_i - e_j = \alpha_i + \dots + \alpha_{j-1}$. A root with $j > r$ would contribute a positive coefficient to some α_k with $k \geq r$. Such contributions cannot cancel in a nonnegative sum of positive roots, while all these coefficients of δ are zero. Thus every root partition of δ uses only roots in Φ_r^+ , so $P_d(\delta) = P_r(\delta)$. $\square$

B Additional cost of an unknown spectrum

We use the notation of Section 6. When the spectrum is unknown, the code's weights P_n must be chosen independently of x . Equation (50) identifies the resulting penalty as the classical relative entropy $D(q||P_n)$; the eigenbasis contribution remains $R_n(x)$. We now evaluate this additional cost for Hayashi's universal codes.

Assume full rank and $x_1 > \dots > x_d > 0$. Write⁴

$$\Delta(x) := \prod_{i < j} (x_i - x_j), \quad v(x) := \sum_{i < j} \frac{x_i \log x_i - x_j \log x_j}{x_i - x_j}.$$

Equation (51) gives $s_{\text{blk}}(x) = v(x) - \log \Delta(x)$, so

$$R_n(x) = \frac{d(d-1)}{2} \log n + 2 \log \Delta(x) - v(x) - \sum_{k=1}^{d-1} \log k! + o(1). \quad (69)$$

⁴Our $v(x)$ is the permutation expression in [37, Eq. (10)], following [41]:

$$v(x) = \frac{1}{\Delta(x)} \sum_{\pi \in S_d} \text{sgn}(\pi) \left(\prod_i x_i^{d-\pi(i)} \right) \log \left(\prod_i x_i^{d-\pi(i)} \right).$$

Indeed, applying $\sum_i x_i \log x_i \partial_{x_i}$ to the product and determinant expansions of $\Delta(x) = \det[x_i^{d-j}]_{i,j=1}^d$ gives the two forms.

Hayashi's uniform code $\sigma_{U,n} = \sigma_{P_{U,n}}$ assigns equal weight to each of the $\sim n^{d-1}/[d!(d-1)!]$ Young diagrams [37]. Define

$$C(x) := 2 \log \Delta(x) - \frac{1}{2} \sum_i \log x_i - v(x). \quad (70)$$

With the Weyl denominator $\prod_{i < j} (j - i) = \prod_{k=1}^{d-1} k!$, he obtained the expansion

$$\begin{aligned} D(\rho^{\otimes n} \| \sigma_{U,n}) &= \frac{d^2 - 1}{2} \log n + C(x) - \frac{d-1}{2} \log(2\pi e) \\ &\quad - \sum_{k=1}^{d-1} \log k! - \log[d!(d-1)!] + o(1). \end{aligned} \quad (71)$$

Subtracting (69) in the exact decomposition (50) gives

$$D(q \| P_{U,n}) = \frac{d-1}{2} \log n - \frac{1}{2} \sum_i \log x_i - \frac{d-1}{2} \log(2\pi e) - \log[d!(d-1)!] + o(1). \quad (72)$$

The extra leading cost is therefore $\frac{d-1}{2} \log n$, as in classical coding with $d-1$ unknown spectral parameters [42]. Its constant contains the multinomial Fisher determinant and the normalization of the uniform spectral prior.

The uniform code is one choice of prior. Optimizing the asymptotic constant over codes for all fixed full-rank states gives Hayashi's minimax value [37]:

$$\begin{aligned} \inf_{\{\sigma_n\}} \sup_{\rho > 0} \lim_{n \rightarrow \infty} \left[D(\rho^{\otimes n} \| \sigma_n) - \frac{d^2 - 1}{2} \log n \right] \\ = -\frac{d-1}{2} \log(2\pi e) - \sum_{k=1}^{d-1} \log k! + \log \int 2^{C(x)} dx. \end{aligned} \quad (73)$$

The infimum is over code sequences for which the displayed limit exists in the extended real numbers for every fixed full-rank state ρ . The integral is over ordered positive spectra with $\sum_i x_i = 1$ and $dx = dx_1 \cdots dx_{d-1}$. An asymptotically optimizing code is the continuous mixture

$$\sigma_{J,n} := \int J(x) \overline{\rho(x)^{\otimes n}} dx, \quad J(x) := \frac{2^{C(x)}}{\int 2^{C(y)} dy},$$

where $\rho(x) = \text{diag}(x)$ and the overline denotes Haar averaging. This is the quantum analogue of the Jeffreys-prior construction [42, 43]. Changing the prior changes the classical term in (50). For both codes, the leading overhead is

$$\frac{d^2 - 1}{2} \log n = \underbrace{\frac{d(d-1)}{2} \log n}_{\text{unknown eigenbasis}} + \underbrace{\frac{d-1}{2} \log n}_{\text{unknown spectrum}}.$$

QMDL determines the first contribution through its additive constant, with the block-entropy correction in (48); universal compression must also pay for the unknown spectrum.

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